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RESEARCH ARTICLE

Performance Analysis of Machine Learning Techniques for Server Health Monitoring Using Time Series Data Against DDoS Attacks

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ABSTRACT In today's digitally driven business environment, the continuous operation of online enterprises relies heavily on resilient server infrastructures. The increasing frequency and sophistication of cyber-attacks, particularly Distributed Denial of Service (DDoS) attacks, pose significant challenges to maintain server health. DDoS attacks are malicious attempts to disrupt normal traffic to a server by overwhelming it with fake requests. Their impact has grown alarmingly, with a 117% year-over-year increase in network-layer DDoS attacks observed in Q4 of 2023. This study introduces a novel two-stage server health monitoring system designed to detect and assess the severity of such attacks. In Stage I: DDoS Attack Detection, binary classification techniques—machine learning methods that divide data into two groups—analyze network traffic to differentiate normal traffic from DDoS attack traffic. The light-gradient boosting model (LGBM) demonstrated exceptional performance on the testing set with an accuracy of 0.9988, AUC of 0.9999, and other metrics such as recall, precision, and F1 score of 0.9988, indicating high reliability. Once a DDoS attack is detected, the system moves to Stage II: Attack Severity Assessment, where a multiclass classification method categorizes the attack's severity as low, moderate, or high. The gradient boosting classifier (GBC) emerged as the best-performing model in this stage, achieving an accuracy of 0.9994, AUC of 1, recall, precision, and F1 score of 0.9994 on the testing set. This research provides a robust approach for detecting attack severity, enabling cybersecurity professionals to allocate resources efficiently and implement timely mitigation to safeguard server health.

INDEX TERMS Cyber security, distributed denial of service attacks, light gradient boosting, machine learning, server health monitoring.

I. INTRODUCTION

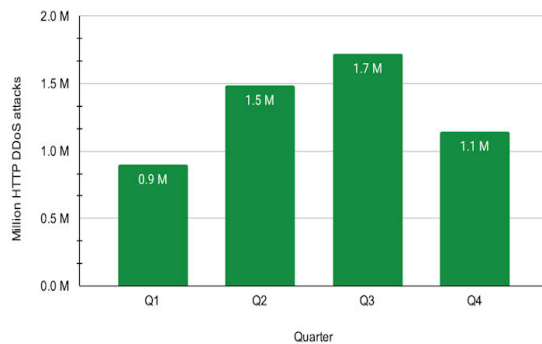
The operational continuity of online business in today's digital age relies heavily on a robust and reliable server infrastructure. However, with the rapid pace of technological advancements, cyber-attacks have become increasingly complex. Among these, Distributed Denial of Service (DDoS) attacks [1] have emerged as formidable adversaries capable of crippling networks and causing significant disruptions across industries. These attacks pose substantial risks, especially in

critical sectors, such as banking, where they can result in severe financial losses.

DDoS attacks are intentional, malicious efforts to overwhelm a server, network, or service by flooding it with an excessive volume of unauthorized traffic, rendering it inaccessible to legitimate users [2]. These attacks exploit vulnerabilities in the network protocols and application layers, amplifying the traffic volume to overwhelm the target systems. The frequency and sophistication of such attacks have escalated significantly, with a 117% increase in network-layer DDoS attacks recorded in Q4 2023 [3]. For instance, HTTP DDoS attacks surged from 0.9 million in Q1 2023 to 1.1 million by Q4 2023, while network DDoS attacks

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HTTP DDoS Attacks in 2023



Network-layer DDoS attacks in 2023

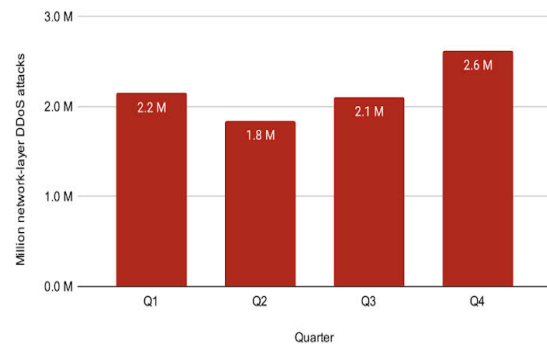


FIGURE 1. 2023 DDoS attacks in numbers.Source: <https://blog.cloudflare.com/ddos-threat-report-2023-q4>.

increased from 2.2 million to 2.6 million over the same period. These statistics [3] highlight the increasing severity of DDoS attacks and underscore the critical need for effective defense mechanisms. Insights from big data analytics further reveal the financial impact of these cyberattacks, emphasizing the importance of proactive mitigation strategies [4].

Despite ongoing efforts [5], [6], [7], [8] to mitigate DDoS attacks, several research gaps remain in the literature. First, many studies rely on outdated datasets that fail to capture the dynamic and evolving nature of the cyberthreat landscape, undermining the reliability of detection methods. Addressing this limitation necessitates the use of contemporary and comprehensive datasets to enhance the relevance and accuracy of the detection techniques. Second, current research predominantly focuses on distinguishing between normal and DDoS attack traffic, with limited attention paid to quantifying the severity levels of such attacks. Understanding the severity spectrum is essential for refining detection systems and prioritizing mitigation strategies effectively.

Finally, there is a notable lack of exploration of the temporal evolution of DDoS attacks. Analyzing time-series data can reveal trends and emerging characteristics, enabling the development of adaptive and resilient detection frameworks. Moreover, incorporating parameter tuning and statistical analytics into QoS optimization has shown promise in improving cyber resilience [9].

The proposed work addresses all these issues by making the following contributions:

1. **Introduction of a Two-stage Server Health Monitoring System:** This paper presents a novel two-stage framework specifically designed to detect and assess the severity of DDoS attacks, ensuring comprehensive server health monitoring.
2. **Binary Classification for DDoS Attack Detection:** Stage I employs machine learning techniques for binary classification to distinguish between normal and DDoS attack traffic, achieving exceptionally high accuracy and reliability, with models such as the Light Gradient Boosting Machine (LGBM) excelling in this stage.
3. **Multiclass Classification for Attack Severity Assessment:** Stage II uses multiclass classification to

categorize the severity of detected DDoS attacks into low, moderate, or high levels. The Gradient Boosting Classifier (GBC) demonstrated near-perfect performance at this stage.

4. **Extensive Use of Diverse Machine Learning Models:** The study employed a wide range of machine learning algorithms to ensure robust detection and severity classification. These include traditional models like Logistic Regression (LR), Decision Tree (DT), and K-Nearest Neighbors (KNN); ensemble-based models such as Light Gradient Boosting Machine (LGBM), Gradient Boosting Classifier (GBC), Adaptive Boosting (ADA), Random Forest (RF), Extra Trees (ET), and Cat Boost; as well as regularization-focused approaches like Ridge Regression (RIDGE). This diverse selection enables comprehensive performance comparison across algorithmic paradigms.
5. **Adaptability and Scalability for Other DDoS Attack Types:** While the system focuses on SYN Flood attacks, the methodology serves as a reference template for addressing other types of DDoS Attacks. By defining specific equations and constants tailored to normal traffic and severity levels for different attack types, the framework can be extended to address reflective- or amplification-based attacks, making it highly versatile and scalable.
6. **Leveraging Specialized Time-Series Data:** This study utilized the TSD-DDOS dataset, specifically designed for DDoS attack detection. This dataset allows for effective analysis of temporal trends in network traffic and supports real-time detection strategies.
7. **Utilization of Advanced Performance Metrics:** The evaluation of model performance incorporates a range of metrics, including accuracy, AUC, precision, recall, F1 score, Kappa, and MCC, ensuring a rigorous assessment of the efficacy of the proposed system.
8. **Enhancement of Server Resilience Against Cyber Attacks:** By offering a robust detection and mitigation mechanism, the proposed system provides network administrators and cybersecurity professionals with

actionable insights into safeguarding server health and maintaining uninterrupted online services.

The remainder of this paper is organized as follows. Section II explores the existing research landscape of DDoS attack detection. Section III describes the proposed method and explains the approach used for detecting and assessing the severity of DDoS attacks. Section IV presents experimental results. Section V provides a detailed discussion and analysis of the findings and their implications. Finally, Section VI concludes the study and outlines directions for future research.

II. RELATED WORKS

In today's digital era, reliable server health monitoring is critical for maintaining uninterrupted online services. DDoS attacks can overwhelm servers and cause major disruptions. Machine learning techniques using time series data offer a promising way to detect these attacks early and assess their severity. This study examines various machine learning methods for DDoS attack detection, evaluating their ability to detect attacks accurately. We review the existing literature by categorizing studies according to benchmark datasets—such as KDD CUP 99, NSL-KDD, UNSW-NB15, CIC-IDS2017, and CIC-IDS2019, which play a crucial role in understanding model performance. This approach highlights the usefulness of these datasets in evaluating and improving detection models.

KDD CUP 99: The KDD CUP 99 [10] is a widely used benchmark dataset for evaluating intrusion detection systems (IDS), particularly for detecting various types of network attacks, including Denial of Service (DoS) attacks. The dataset contains labeled records representing different network traffic with DoS attack types, such as smurf, neptune (SYN Flood), teardrop, back, pod (ping of death), and land. These records allow researchers to develop and test IDS models that can identify and mitigate DoS attacks by analyzing patterns in network traffic that are indicative of malicious activities.

The multi-classifier system proposed by W.S. and B. [11] uses four different models: AdaBoost, support vector machines (SVM), linear discriminant analysis (LDA), and random forest (RF). Their system achieved exceptional performance on the KDD CUP 99 dataset, with a sensitivity of 99.75% and specificity of 95.69%, which is significantly higher than those of other comparable classifiers. This suggests that the proposed system is highly effective in identifying DDoS attacks.

Using principal component analysis (PCA) and the random forest classification technique, Waskle et al. [12] developed a new method for creating an intrusion detection system (IDS) that works well. The researchers used PCA to reduce the dimensionality of the dataset and employed a Random Forest for accurate classification. The study found that the proposed strategy performed better than existing methods such as support vector machines (SVM), naive Bayes, and decision trees in terms of accuracy. In addition, the system

exhibited an impressive performance time of 3.24 minutes and an accuracy rate of 96.78%, demonstrating its efficiency and effectiveness on the KDD CUP 99 dataset.

Fang et al. [13] developed a new method for addressing concerns regarding intrusion detection in computer systems by combining Elman neural networks with strong Support Vector Machine (SVM) methods. The proposed method achieved an outstanding intrusion detection rate of 87.3% with a false alarm rate of zero on the KDD CUP 99 dataset, while maintaining a low false alarm rate of 2.8%. The findings of this study indicate that the use of Elman neural networks and a robust SVM for noise reduction in data can be an effective solution for ensuring the security of information systems. The authors proposed a method that is both reliable and efficient for detecting intrusions in information systems.

Farahnakian and Heikkonen [14] presented a deep autoencoder approach for improving intrusion detection systems. The proposed method is a deep autoencoder-based intrusion-detection system (DAE-IDS). They trained four autoencoders using a greedy, unsupervised, and layer-wise method. Using the KDD CUP 99 dataset, the approach achieved a detection accuracy of 94.71% on the test dataset.

Gupta et al. [15] looked at a deep learning-based approach for an intrusion detection system that uses a stacked autoencoder to find complex patterns and make the best latent representation of the inputs. They trained the proposed model on single-labeled instances from the KDD CUP 99 dataset and standardized the inputs using batch normalization. The results show that the proposed method outperforms all the other algorithms, with an accuracy of 98.17% and a false alarm rate of 0.38%.

Kishore et al. [16] proposed a study that reviewed different types of intrusion detection techniques and demonstrated the effectiveness of deep neural networks in comparison with traditional machine learning techniques. The study utilized a learning rate of 0.1 and executed 1000 iterations while training the model on the KDD CUP 99 dataset. The results of the study indicated that the Random Forest algorithm outperformed the other techniques, achieving an accuracy of 0.927, precision of 0.998, recall of 0.911, and F1-score of 0.951.

Lateef et al. [17] proposed an intrusion detection system (IDS) that utilizes a combination of a recurrent neural network (RNN) and crowd swarm optimization (CSO) algorithms to improve its performance. The KDD CUP 99 dataset was used to evaluate the IDS binary-classification design. The findings of this study show that implementing the CSO algorithm for feature selection and reduction significantly improves the accuracy of the system, resulting in an accuracy rate of 98.34% with only three selected features.

Yang et al. [18] proposed an intrusion detection method using an attention-long short-term memory (LSTM) neural network. This method utilizes the LSTM attention mechanism and memory function to learn the time-series data. They tested the proposed method on the KDD CUP 99 dataset and found it to be efficient, improve accuracy and precision,

and decrease false alarm rates. It provides a foundation and technical support for future intrusion detection technologies.

Muhammad et al. [19] proposed an IDS using a stacked autoencoder and a DNN for feature extraction. They tested the system on the KDD CUP 99, NSL-KDD, and Aegean Wi-Fi datasets, achieving 94.2%, 99.7%, and 99.9% accuracy, respectively, for multiclass classification.

The KDD CUP 99 dataset, which is widely used for evaluating intrusion detection systems (IDS), has several well-documented limitations [20]. These include a significant number of redundant records, which can lead to biased model training and overfitting. A severe class imbalance plagues the dataset and the overrepresentation of certain attack types of skew performance metrics. Furthermore, traffic data reflects the late 1990s network environments, making it less relevant for modern network conditions with more sophisticated attack vectors. Furthermore, generating a dataset in a controlled environment may not accurately represent real-world network traffic, and the lack of diversity in modern attack types limits its usefulness for evaluating contemporary IDS.

NSL-KDD: The NSL-KDD [21] dataset is an improved version of the KDD CUP 99 dataset [15], created to address some of the inherent issues in the original dataset, such as redundancy and imbalance. Researchers commonly use it to evaluate the performance of intrusion detection systems, provide a more balanced data distribution, and eliminate duplicate records to create a more realistic benchmark for model testing.

Amaran and Mohan [22] proposed an intrusion detection system (IDS) that leverages the power of support vector machines (SVM) by optimizing the kernel function selection. They suggested using a Whale Optimization Algorithm (WOA) to select the optimal kernel function for the SVM model. This significantly improves intrusion detection. They evaluated the proposed method on the benchmark NSL-KDD dataset and achieved outstanding results, with an accuracy of 94.09% and a detection rate of 95.02%. These results demonstrate the effectiveness of the proposed OSVM-based IDS in identifying intrusions.

Singhal et al. [23] presented a comprehensive study of various machine-learning approaches for training and testing network intrusion detection models (NIDM). They proposed the use of decision trees, naive Bayes, k-nearest neighbors (KNN), and Support Vector Machine (SVM) algorithms for NIDM. When evaluated individually on the NSL-KDD dataset, these algorithms achieved high accuracy rates, with decision trees achieving 98.088%, Naive Bayes achieving 82.971%, KNN achieving 95.75%, and SVM achieving 81.97%. This study highlights the effectiveness of these machine-learning techniques in identifying and preventing network intrusions.

Liu et al. [24] proposed a new way to divide objects into two groups: normal and attack. It uses k-means and Random Forest (RF) algorithms, along with distributed computing

on the Spark platform, to identify both types of events. The model employs advanced techniques such as convolutional neural networks (CNN) and long short-term memory (LSTM) to classify the abnormal occurrences. To address the issue of unbalanced datasets, the model applies adaptive synthetic sampling (ADASYN). They carefully tested the proposed model on the NSL-KDD and CIS-IDS2017 datasets, and obtained excellent results. The NSL-KDD dataset had an 85.24% multi-target classification accuracy, and the CIS-IDS2017 dataset had an impressive 99.91% accuracy, demonstrating how well it can find and stop network intrusions.

Rokade and Sharma [25] proposed an SVM-IDS technique utilizing deep learning. They tested the method on a NSL-KDD simulated intrusion dataset. They then compared it with several other deep learning methods for network intrusion detection, including the Naive Bayes Artificial Neural Network (NB ANN), Random Forest (RF), and Support Vector Machine (SVM). They discovered that using an efficient rule structure in their proposed technique resulted in improved classification and high-class detection. This machine learning approach yielded an overall system performance of approximately 95%.

Rajadurai and Gandhi [26] presented a novel approach to intrusion detection by proposing the use of stack ensemble learning, which combines the predictions of multiple machine-learning algorithms to improve performance. Using the NSL-KDD dataset, they tested the effectiveness of the stacked ensemble system and discovered that it was much better than other intrusion detection methods, with a 91.06% success rate. Additionally, the study conducted a comprehensive comparison of the proposed system with various commonly used machine learning algorithms, demonstrating its superior performance in this domain.

Haggag et al. [27] proposed an intrusion detection system (IDS) that uses multilayer perceptrons (MLPs) and instance-based learning algorithms to sort and identify possible threats in a computer system. The proposed IDS achieved exceptional performance results on the NSL-KDD dataset, with accuracy rates of 99.87% and 99.82% when using only five and three features, respectively, of the 78 available features for IBK. This study not only demonstrates the efficacy of the proposed IDS but also highlights the importance of feature selection and dimensionality reduction in achieving high-performance intrusion detection systems. The results of this study indicate that the proposed IDS is an effective and efficient approach for identifying and classifying threats in information systems.

To intrusion detection systems (IDS), Pakanzad et al. [28] suggested a mix of convolutional neural networks (CNN) and long short-term memory networks (LSTM) for intrusion detection systems. This method aims to accurately categorize malicious traffic into groups according to attack type. It worked on the NSL-KDD and CICIDS2017 datasets,

showing 98.1 and 96.7 percent accuracy, respectively, for multiple classifications.

Rezaeipanah et al. [29] proposed a method combining deep learning and observer learning to improve intrusion detection in computer networks. This method uses a supervised learning algorithm and deep neural network to optimize the number of hidden layers and neurons in each layer. They tested the proposed method on the NSL-KDD dataset and achieved an accuracy of 97.64%, thus outperforming the MARS and DLNN methods.

A new method proposed by Ahsan et al. [30] suggests the use of a combination of CNN and LSTM to improve intrusion detection. Their results showed that it worked 99.70% of the time on the NSL-KDD dataset, and they obtained good scores for precision, false positives, F1 score, and recall, making it an excellent choice for use in real network infrastructure.

Boukhalfa et al. [31] proposed a new idea for a Network Intrusion Detection System (NIDS) based on Long Short-Term Memory (LSTM) to detect threats and obtain their long-term memories to prevent similar attacks. When tested on the NSL-KDD dataset, the proposed model performed well with an accuracy of 99.98% for two classes and 99.93% for multiple classes. It also had a low false-positive rate of 0.068% for two classes and 0.023% for multiple classes.

Network intrusion detection research widely uses the NSL-KDD dataset, which is an updated version of KDD99; however, it has notable limitations. It suffers from significant class imbalance, leading to biased models, and still contains some duplicate and irrelevant records, risking overfitting. The dataset is outdated, reflecting 1990s network traffic, making it less relevant to modern threats. Its controlled artificial nature further limits the realism and generalizability of the trained models. These issues, along with reliance on manual feature engineering, challenge the development of effective intrusion-detection systems using NSL-KDD.

CIC-IDS2017: The CIC-IDS2017 dataset [32], created by the Canadian Institute for Cybersecurity, is a widely used resource for advancing intrusion detection systems (IDS) research. Designed to address the limitations of older datasets like KDD99 and NSL-KDD, it offers a realistic representation of modern network traffic and security threats. Generated through controlled experiments, the dataset includes a diverse range of attack vectors, such as Denial of Service (DoS) and brute-force attacks, with detailed labeling and extensive network features. Its balance of complexity and comprehensiveness has established CIC-IDS2017 as a benchmark in cybersecurity research, facilitating the development and evaluation of effective IDS techniques.

Hussein Ridha Sayegh et al. [33] suggested a better intrusion detection system (IDS) that uses long short-term memory (LSTM), feature selection, and SMOTE to address the problem of uneven data in IoT networks. Hussein Ridha Sayegh et al. created an LSTM model to capture changes in network traffic over time. They also used SMOTE to

create fake samples for minority groups, which improved the model at identifying rare attack patterns. Experiments on the CICIDS2017, NSL-KDD, and UNSW-NB15 datasets showed that the proposed model performed well with accuracy rates of 99.34%, 99.75%, and 98.31%, respectively. The model also maintained high precision and recall across all the datasets, demonstrating its robustness and suitability for intrusion detection in complex IoT environments.

Rodríguez et al. [34] suggested an Attentive Transformer deep-learning algorithm for identifying intrusions in IoT systems. It includes an automatic, explainable feature selection mechanism that makes the model easier to understand and better at its job. This model employs transformer architecture to focus on the most critical features in a dataset, and automatically selects them to improve the accuracy and transparency of intrusion detection. They implemented the model using the TabNet algorithm, which optimizes tabular data and provides both local and global feature explanations. The model was tested on three widely used datasets: CIC-IDS2017, CSE-CICIDS2018, and CIC-DDoS2019. The results showed that the model was very good at identifying network intrusions, with 97.03% accuracy for CIC-IDS2017, 95.58% accuracy for CSE-CICIDS2018, and 98.51% accuracy for CIC-DDoS2019. In addition, the model provides clear explanations for its predictions, enhancing its applicability in real-world IoT security scenarios.

Tapu et al. [35] suggested a mixed meta-deep learning method for identifying malicious data in packet data networks. This method combines Siamese and prototypical networks to improve classification accuracy with less data. It uses both Siamese networks for binary classification, and prototypical networks for multiclass classification. This enables the sorting of malicious network traffic into various types of attacks. This study used the CSE-CIC-IDS2017 and CSE-CIC-IDS2018 datasets to demonstrate that the model could classify data accurately (over 90% for both binary and multiclass classifications with only 3000 samples). The results indicated the robustness of the model in detecting and classifying network intrusions, even in scenarios with limited training data. Furthermore, the proposed approach addresses the challenges of data scarcity and unbalanced datasets, thereby rendering it highly applicable to real-world network security applications.

A group of researchers led by Halbouni et al. [36] created a convolutional neural network-based intrusion detection system (CNN-IDS) to improve network security by accurately finding and categorizing break-ins. They evaluated the system using the CIC-IDS2017 dataset and achieved a remarkable detection rate of 99.55% and a false alarm rate (FAR) of 0.12% in multiclass classification. This model addresses the need for low false alarms and high detection rates in modern network security systems, making it highly effective in protecting against various cyberthreats.

The CIC-IDS 2017 dataset, which has been widely adopted for intrusion detection research, has several critical

limitations. These include a lack of information on attack severity, which hinders the understanding of the impact of different threats and issues with data imbalance and potential data leakage, leading to overfitting and inflated accuracy scores. Additionally, the dataset's controlled environment and outdated nature do not reflect modern network traffic or emerging threats, reducing its applicability in real-world scenarios. Furthermore, Bertoli et al. [37] highlighted that the aging of the dataset and lack of a systematic update approach contributed to its limited relevance to contemporary network architectures. Dube et al. [38] also demonstrated the importance of having a well-planned experiment and properly preparing the data before drawing any conclusions. They also advocated balanced class distribution and meticulous selection of performance metrics to ensure robust results applicable to various scenarios. Therefore, to ensure its effectiveness in modern intrusion detection research, we must carefully handle the CIC-IDS 2017 and ideally supplement it with up-to-date data.

UNSW-NB15: The UNSW-NB15 dataset [39] is a widely used benchmark for evaluating network intrusion detection systems (NIDS). The Australian Center for Cyber Security (ACCS) created this dataset in 2015 using the IXIA PerfectStorm tool to simulate modern network traffic, which encompasses a wide range of attack types, such as DoS, backdoors, fuzzers, and analytic attacks. With more than two million records, the dataset contains nine different attack types and normal traffic. It contains 49 features derived from packet-level network traffic data and ground truth labels for supervised learning tasks. The realistic and diverse representation of contemporary network traffic in UNSW-NB15 makes it a crucial resource for developing and testing NIDS models [40].

On the UNSW-NB15 dataset, Hammad et al. [41] suggested using Naive Bayes (NB), Random Forest (RF), J48, and ZeroR, among other machine learning algorithms, for classification. The findings of this study revealed that RF performed exceptionally well, with an accuracy of 97.59%, whereas J48 also performed well, with an accuracy of 93.78%. Additionally, the researchers applied K-means and expectation maximization (EM) clustering algorithms to the dataset, providing a comprehensive evaluation of various classification techniques.

Siddiqi and Pak [42] suggested a flexible method to improve network intrusion detection based on machine learning by identifying effective single and hybrid normalization methods. The study used Support Vector Machines (SVM), k-nearest neighbors (k-NN), Decision Trees, and Random Forest algorithms evaluated on the UNSW-NB15 and NSL-KDD datasets. Hybrid normalization techniques significantly improved the accuracy of the model. However, reliance on the UNSW-NB15 dataset, which has some known limitations, such as class imbalance and controlled environment traffic, may influence the generalizability of the results to more diverse real-world scenarios.

Using convolutional neural networks (CNN) and long short-term memory (LSTM) networks, Zeeshan et al. [43] developed a protocol-based deep-learning method to detect DoS and DDoS attacks. Researchers conducted a study using the UNSW-NB15 and Bot-IoT datasets, achieving high accuracy, with the CNN model achieving 98.2% accuracy on UNSW-NB15. Despite these promising results, it is crucial to acknowledge that the UNSW-NB15 dataset exhibits some features of redundancy and data duplication, potentially compromising the perceived robustness of the model.

Panda et al. [44] used decision trees, random forests, and gradient boosting on the UNSW-NB15 and Bot-IoT datasets to create a feature engineering and machine learning model that is effective for finding IoT-botnet cyberattacks. The feature engineering approach significantly enhanced the performance of the model. However, given that the UNSW-NB15 dataset has certain limitations such as limited attack diversity and synthetic traffic, the applicability of the results in more complex real-world environments may require further investigation.

Ozkan-Okay et al. [45] discussed the SABADT system, which is a hybrid intrusion detection method for WLANs that uses both signature-based and algorithm-based methods to identify threats. The study used SVM, Naive Bayes, and k-NN algorithms evaluated on the UNSW-NB15 and NSL-KDD datasets. The hybrid approach yielded significant improvements in the detection accuracy; however, it is worth noting that the UNSW-NB15 dataset's class imbalance and its non-representative traffic patterns might influence the system's effectiveness in broader real-world scenarios.

Das et al. [46] proposed a novel entropy-based feature-encoding scheme to enhance machine-learning-based malware detection. They evaluated the scheme using the KDD CUP 99, UNSW-NB15, and CIC-Evasive-PDFMal2022 datasets. The ensemble classifier trained with the proposed encoding achieved F1 scores of 99.99% for KDD CUP 99 and 99.27% for CIC-Evasive-PDFMal2022. However, because the UNSW-NB15 dataset has some problems such as class imbalance and duplicate features, the model might not work as well in real-life situations that are more changing.

CIC-IDS2019: The CIC-DDoS2019 dataset [47] is a widely used benchmark for evaluating machine learning (ML) and deep learning (DL) models in DDoS attack detection owing to its comprehensive coverage of real-world attack scenarios. Developed by the Canadian Institute for Cybersecurity (CIC), it includes various attack types, such as TCP, UDP, and reflection-based attacks, making it essential for intrusion detection research. Studies have shown that different learning approaches, feature selection techniques, and explainability models enhance DDoS detection across SDN, IoT, Industry 5.0, and federated learning environments. The rich attack diversity of the dataset continues to support advancements in real-world cybersecurity solutions.

Several studies have leveraged CIC-DDoS2019 to enhance DDoS detection models using advanced techniques such as

ensemble learning, deep learning, federated learning, and explainability frameworks. Alashhab et al. [48] introduced an ensemble online machine-learning model for SDN-based DDoS detection. The model dynamically updates attack signatures through online learning, achieving a 99.2% detection rate and significantly improving adaptability to evolving threats. Mahadik et al. [49] developed an Edge-Federated Learning-Based Intrusion Detection System (Edge-FL-IDS) for heterogeneous IoT environments. The federated learning (FL) model trained on CIC-DDoS2019 achieved an accuracy of up to 99.98%, demonstrating the potential of decentralized learning in securing IoT systems.

Ferrag et al. [50] developed a deep learning-based Intrusion Detection System (IDS) for DDoS attacks in Agriculture 4.0 using RNN, CNN, and DNN models. Their system was tested on the TON_IoT and CIC-DDoS2019 datasets to evaluate the performance in both multiclass and binary classifications. Aldhyani and Alkahtani [51] developed a deep learning-based Intrusion Detection System (IDS) to protect Agriculture 4.0 networks from DDoS attacks. Using the CICDDoS2019 dataset, they trained CNN-LSTM and LSTM models, achieving 100% accuracy in detecting threats. Their work highlights the role of deep learning in protecting smart agriculture from cyberattacks. Assadhan et al. [52] proposed an LSTM-based IDS for DDoS attack classification, integrating SHAPPD to improve interpretability. Based on CICDDoS2019, the model accurately classifies 12 attack types, with SHAPPD providing better insights into LSTM decisions by analyzing feature dependencies and sequential packet patterns. In the industrial domain, Javeed et al. [53] proposed an explainable and resilient IDS for Industry 5.0, combining BiLSTM, Bi-GRU, and SHAP to improve intrusion detection and interpretability. Tested on CICDDoS2019, it achieved 99.77% accuracy, outperforming existing models and enhancing cybersecurity in industrial networks.

Fotse et al. [54] introduced FedLAD, a multi-controller federated learning model for large-scale SDN environments. The system, trained on the CICDDoS2019, CIC-DoS2017, and InSDN datasets, achieved an accuracy of over 98% while minimizing resource consumption. Saveetha et al. [55] proposed a federated machine learning and blockchain framework for DDoS attack detection, integrating a reputation-based miner selection to prevent model poisoning. Using CICDDoS2019 and IDS2018 datasets, Random Forest achieved 99.1% accuracy, outperforming traditional methods. The framework enhances security, reliability, and model integrity in decentralized networks. Shieh et al. [56] proposed CNN-RPL, an Intrusion Detection System (IDS) that combines Convolutional Neural Networks (CNN) with Reciprocal Points Learning (RPL) to detect both known and unknown DDoS attacks. Tested on CICIDS2017 and CICDDoS2019, it achieved 99.93% accuracy for known attacks and 98.51% for unknown attacks, improving detection efficiency while reducing model complexity.

Research Gaps: While existing literature presents a plethora of approaches and techniques for DDoS attack detection utilizing various machine learning algorithms, there are notable research gaps that warrant further investigation.

- **Dataset Relevance and Quality:** Many studies rely on outdated datasets such as NSL KDD and KDD-Cup'99, which have been extensively scrutinized for their limitations and biases. However, these datasets do not adequately reflect the current landscape of network traffic and may not encompass a diverse range of contemporary cyber threats. Furthermore, there is a lack of exploration of the issues and biases inherent in these datasets, potentially affecting the generalizability and reliability of the proposed detection models.
- **Attack Severity Analysis:** Existing studies on benchmark datasets such as UNSW-NB15, CIC-IDS2017, and CIC-IDS2019 primarily focus on using machine learning techniques to identify DDoS attacks, without exploring the different severity levels these attacks may exhibit. A thorough examination of attack severity provides valuable insights into the sophistication of DDoS attacks and aids in developing more effective mitigation strategies.
- **Temporal Dynamics of DDoS Attacks:** Time-series analysis of DDoS attacks is crucial for capturing the temporal dynamics and evolution of attack patterns over time. However, there is a lack of emphasis on leveraging time-series datasets to understand the temporal aspects of DDoS attacks, which can reveal valuable patterns and trends essential for proactive threat mitigation strategies.

III. PROPOSED METHOD

Distributed Denial of Service (DDoS) attacks pose significant threats to server infrastructures. Attackers use varying speeds of abnormal traffic to compromise system availability and performance. The severity and impact of these attacks are closely linked to the rate at which abnormal traffic is generated. Consequently, accurately detecting and assessing the severity of DDoS attacks is essential for deploying effective mitigation strategies and protecting server health.

This study presents an innovative two-stage server health monitoring system developed to detect and assess the severity of DDoS attacks as illustrated in Figure 2. The first stage, “DDoS Attack Detection,” involves the analysis of network traffic data through a binary classification technique. This stage effectively distinguishes between normal traffic and DDoS attack traffic, allowing the system to detect the presence of DDoS malicious activity.

Upon detection of an attack, the system advances to Stage II: “Attack Severity Assessment” utilizes a multi-class classification technique to evaluate the severity of the detected DDoS attacks, classifying them into three categories based on their intensity: low, moderate, or high as detailed in Table 1. The effectiveness of the proposed method was validated through testing on the benchmark

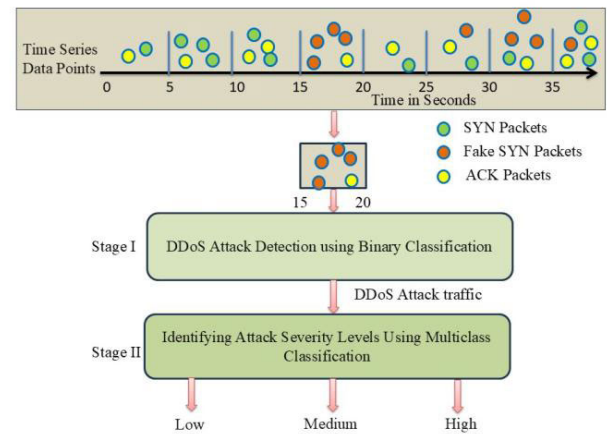
TABLE 1. Attack severity levels.

S No	Attack Severity	Sever health Status	Server availability and Performance
1	No	Good	Response time is stable and within normal limits. The server is operational and has sufficient resources. Performance indicators are excellent.
2	Low	Fair	Response time is stable and within normal limits. The server is operational and has adequate resources. Performance indicators are favorable.
3	Moderate	Serious	Response times are unstable and do not meet normal standards. Although the server remains operational, it sometimes lacks sufficient resources, resulting in questionable performance indicators.
4	High	Critical	Response times are unstable and exceed normal limits. The server is down due to insufficient resources, and performance indicators are poor.

TSD-DDoS dataset [53]. This dataset is commonly used to identify various types of TCP-based flooding attacks, including TCP-SYN, TCP-SYN-ACK, TCP-ACK, and TCP-RST attacks. The proposed method is specifically optimized for the detection of TCP-SYN flooding attack, which is widely recognized as a common form of DDoS attack targeting server infrastructures.

A. TCP-SYN FLOODING ATTACK

Services provided on the Internet, such as the World Wide Web (HTTP), e-mail (SMTP/IMAP), File Transfer Protocol (FTP), and remote terminal access (Telnet), all operate on top of the Transmission Control Protocol (TCP). TCP is a reliable, connection-oriented transport-layer protocol that ensures data is transmitted accurately and in the correct order between communicating systems. When users request these services, TCP initiates the process of establishing a connection between the user's computer (the client) and the server hosting the service. This process guarantees that both systems are ready to exchange data by following a precise handshake procedure, which ensures reliability, congestion control, and data flow management. The process of

**FIGURE 2.** Two-stage server health monitoring system.

connection establishment, as illustrated in Figure 3, enables smooth, error-free communication necessary for services like web browsing, email communication, file transfers, and remote access to servers.

In a typical TCP connection establishment process, the user's computer initiates the connection by sending an SYN packet to the server. In response, the server sends a SYN-ACK packet to the user's computer, acknowledging the initial SYN packet. Upon receiving the SYN-ACK packet, the user's computer sends an ACK packet back to the server, confirming the receipt of the SYN-ACK. With this three-way handshake complete, the TCP connection is established. Once the connection is open, the server provides the requested service, leading to the exchange of data packets between the two computers. Finally, the connection is terminated by closing the communication link.

Each TCP connection is uniquely identified by four parameters: the User Computer IP address, Port Number, Server Computer IP address, and Server Port Number. The server computer maintains dedicated data structures to manage the information associated with each TCP connection. These data structures are instantiated upon receipt of an SYN packet from the user, whereupon the server responds with an SYN-ACK and awaits an ACK packet from the User Computer within a predefined timeframe, typically 75 s.

This transitional phase is referred to as a "half-open connection" or "SYN Received" state. In routine scenarios, the expected ACK packet is promptly received, and all concurrent half-open connections are duly managed within the backlog queue of the operating system.

During a TCP SYN Flood attack, an attacker overwhelms the server by sending fake SYN packets. These packets, crafted with unreachable (disconnected) computer IP addresses, were designed to overwhelm the capacity of the server to process connection requests. Consequently, the server responds diligently with SYN-ACK packets, awaiting the final ACK packets to complete the three-way connection. However, because spoofed SYN-ACK packets fail to reach

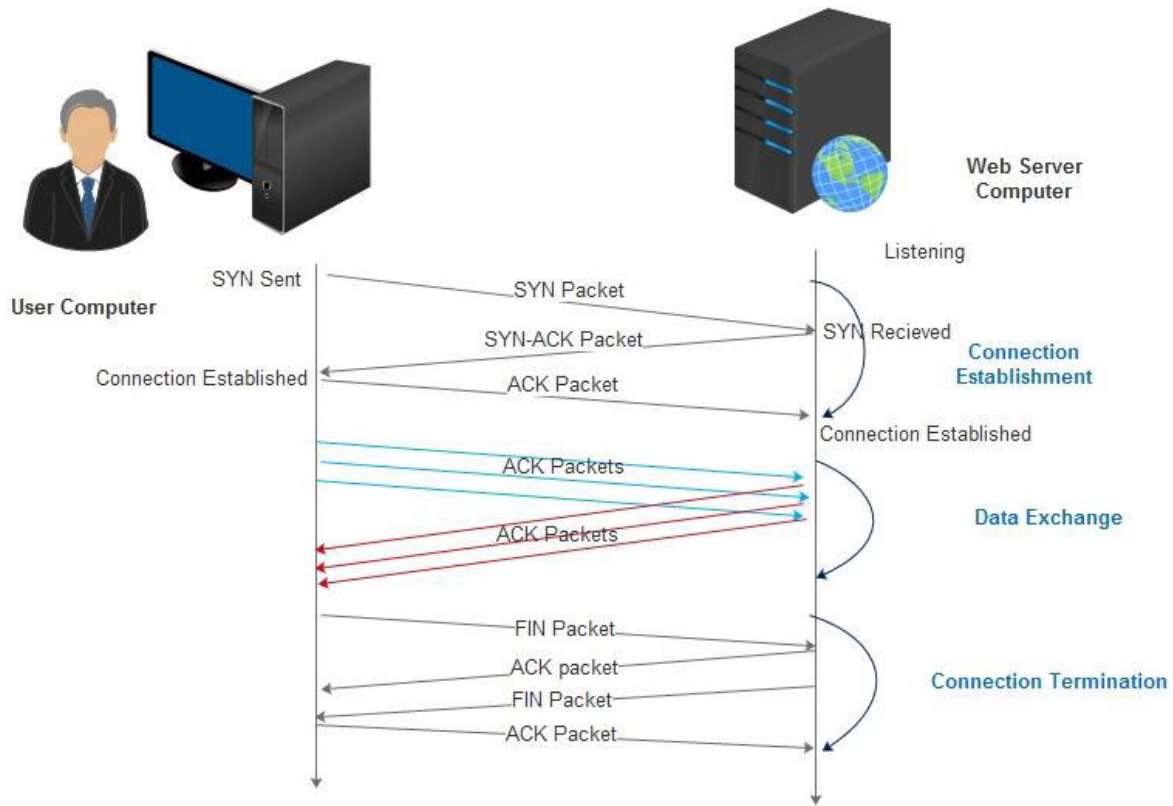


FIGURE 3. TCP connection stages.

any legitimate machine in the network, the server remains indefinitely awaiting ACK responses. In a TCP-SYN flooding DDoS scenario, the attacker orchestrates the flooding of millions of forged SYN packets, thereby saturating the backlog queue of the designated port of the server with numerous half-open connections. Consequently, incoming connection requests are systematically ignored, leading to denial of service from the server's end.

The proposed system targets the identification of TCP-SYN flooding DDoS attacks by analyzing network traffic as a time series and applying classification techniques. It employs binary classification to detect the presence of an attack and multiclass classification to assess its severity, following the steps outlined in Figure 4.

1) TIME WINDOWING

The Time-Series Dataset for DDoS (TSD-DDoS) [57] was created by replaying CIC-DDoS2019 [47] pcap files at network speeds of up to 20 Gbps using *tcpreplay* [58], with *Wireshark* [59] capturing traffic at 5-second intervals. The dataset comprises network traffic data segmented into fixed-size windows, each spanning 5 seconds. Each window serves as a distinct instance or sample for classification, facilitating time-series analysis of DDoS attack patterns.

2) FEATURE EXTRACTION

The feature extraction process focuses on deriving relevant features from network traffic each time window to effectively

capture temporal patterns. In the proposed method, the number of SYN and ACK packets is selected as the key feature from the TSD-DDoS dataset. These features are crucial for representing time-series characteristics associated with TCP SYN Flood attacks. They provided valuable insights into network traffic behavior during each interval, highlighting patterns linked to SYN and ACK packets. This information plays a vital role in detecting potential SYN flood DDoS attacks and assessing their severities.

3) LABELING

In the labeling stage, feature scores were calculated for the selected features. These scores were calculated from the traffic captured in each 5-second window. The scores were then used to assign the class labels. The labeling process followed the criteria outlined in Equations (1), (2), and (3). Equation (1) assesses server health, Equation (2) labels traffic as normal or indicative of a DDoS attack, and Equation (3) further categorizes the traffic into low, moderate, or high severity levels. This systematic approach ensured that server health in each time window was accurately labeled, enabling the effective differentiation and classification of network traffic patterns for subsequent analysis.

Let $P_i = \{P_{i1}, P_{i2}, P_{i3}, P_{i4} \dots P_{in}\}$ is the set of n TCP packets in the i^{th} period.

The health of the server in the i^{th} time period is defined as

$$H = \sum_{k=1}^n f_{\text{SYN}}(P_{ik}) - \sum_{k=1}^n f_{\text{ACK}}(P_{ik}), \quad (1)$$

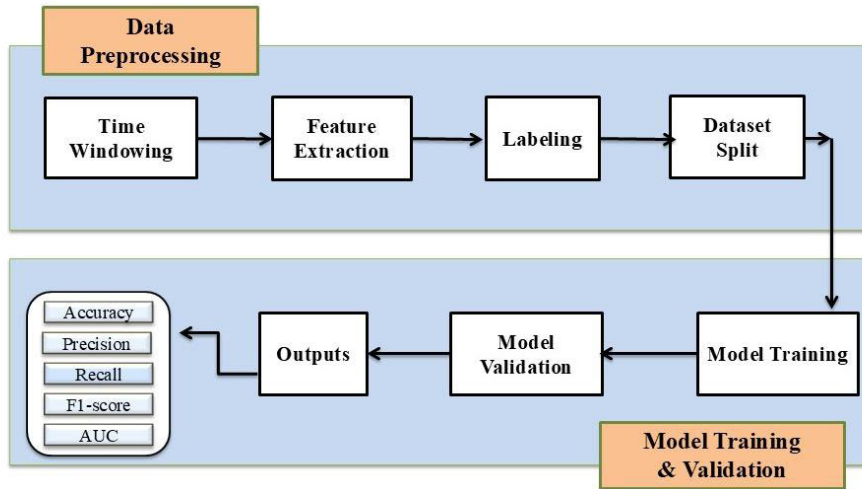


FIGURE 4. DDoS attack detection and severity classification process.

where

$$f_{SYN}(P_{ik}) = \begin{cases} 1 & \text{if } P'_{ik} \text{'s TCP SYN Flag is set} \\ 0 & \text{if } P'_{ik} \text{'s TCP SYN Flag is not set} \end{cases}$$

$$f_{ACK}(P_{ik}) = \begin{cases} 1 & \text{if } P'_{ik} \text{'s TCP ACKFlag is set} \\ 0 & \text{if } P'_{ik} \text{'s TCP ACKFlag is not set} \end{cases}$$

$$h(H) = \begin{cases} \text{Normal,} & \text{if } H < 0 \\ \text{DDoS Attack traffic,} & \text{if } H > 0 \end{cases} \quad (2)$$

$$h(H) = \begin{cases} \text{Low Severity,} & \text{if } H > 0 \text{ and } H < 10^a \\ \text{Moderate Severity,} & \text{if } H > 100 \text{ and } H < 10^b \\ \text{High Severity,} & \text{if } H > 10^c \end{cases} \quad (3)$$

4) DATASET SPLIT

This study utilizes the TSD-DDoS dataset, derived from the CIC-DDoS2019 benchmark dataset, to improve DDoS attack detection and severity assessment through time-series analysis. The dataset includes two CSV files, as shown in Table 2, containing features such as the pcap file number, time period serial number, and counts of SYN, SYN-ACK, ACK, RESET, and TCP packets. This dataset provides a modern time-series perspective for developing advanced DDoS detection methods, making it a valuable tool for improving cybersecurity resilience.

The training set used in this study contained 5,531 samples and the testing set included 5,892 samples. Figures 5 and 6 present detailed visualizations of the time-series samples.

5) CLASSIFICATION MODEL TRAINING

The machine learning models used in this study represent a mix of traditional, ensemble-based, and regularization-focused approaches, enabling the exploration of their comparative performance, adaptability to dataset characteristics, and computational efficiency. The choice of these machine learning models is driven by the need for a

TABLE 2. Dataset details.

Data set	Size
Training Set (jan-12-dataset-final-version-csv-ieee-dataport.csv) [57]	5531 samples [Figure 5]
Testing Set (mar-11-dataset-final-version-csv-ieee-dataport.csv) [57]	5892 samples [Figure 6]

comprehensive evaluation of performance across different algorithmic paradigms.

Ensemble models such as the Light Gradient Boosting Machine (LGBM), Categorical Boosting (CatBoost), Random Forest (RF), Gradient Boosting Classifier (GBC), Extra Trees (ET), and AdaBoost (ADA) enhance prediction accuracy by combining weak learners. LightGBM and CatBoost are highly scalable for large datasets, whereas RF and ET handle noisy data effectively. ADA focuses on misclassified instances, improving the detection of subtle attack patterns.

Tree-based models, such as decision trees (DT), offer interpretability and handle both numerical and categorical features efficiently.

Distance-based models, such as K Nearest Neighbors (KNN), classify traffic based on similarity, making them effective for distinguishing legitimate vs. malicious traffic.

Linear models, including Ridge Classifier (RIDGE) and Linear Discriminant Analysis (LDA), are computationally efficient, with RIDGE preventing overfitting and LDA maximizing class separability. Probabilistic models such as Quadratic Discriminant Analysis (QDA) adapt to varying class variances, enabling more accurate predictions in diverse traffic conditions. Logistic Regression (LR) serves as a baseline due to its simplicity and interpretability.

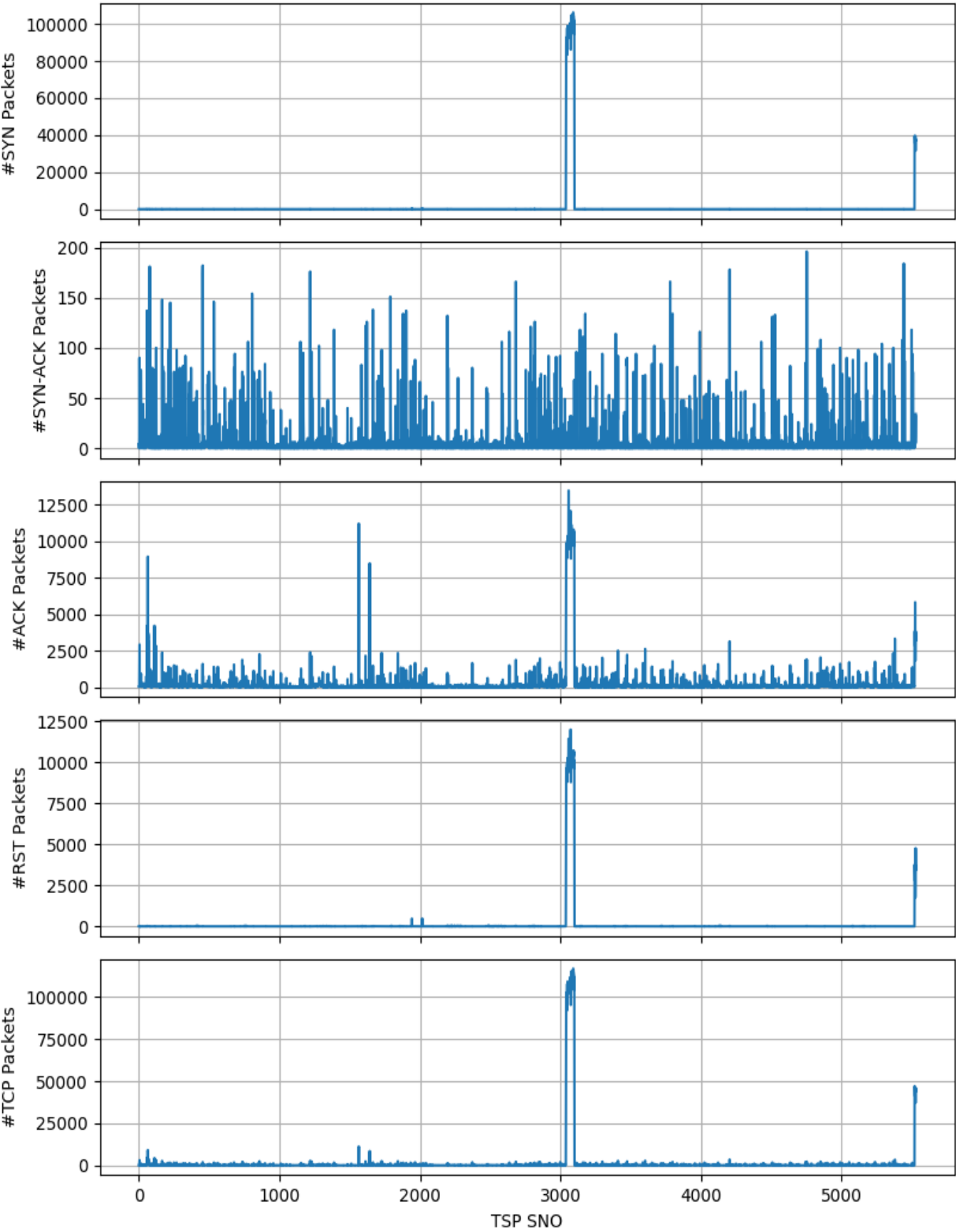


FIGURE 5. Training set samples.

This diverse selection ensures a well-rounded comparison of model performance, optimizing DDoS detection and severity classification across different network conditions.

6) MODEL EVALUATION

Stage I employed binary classification models for DDoS attack detection, while Stage II utilized multiclass

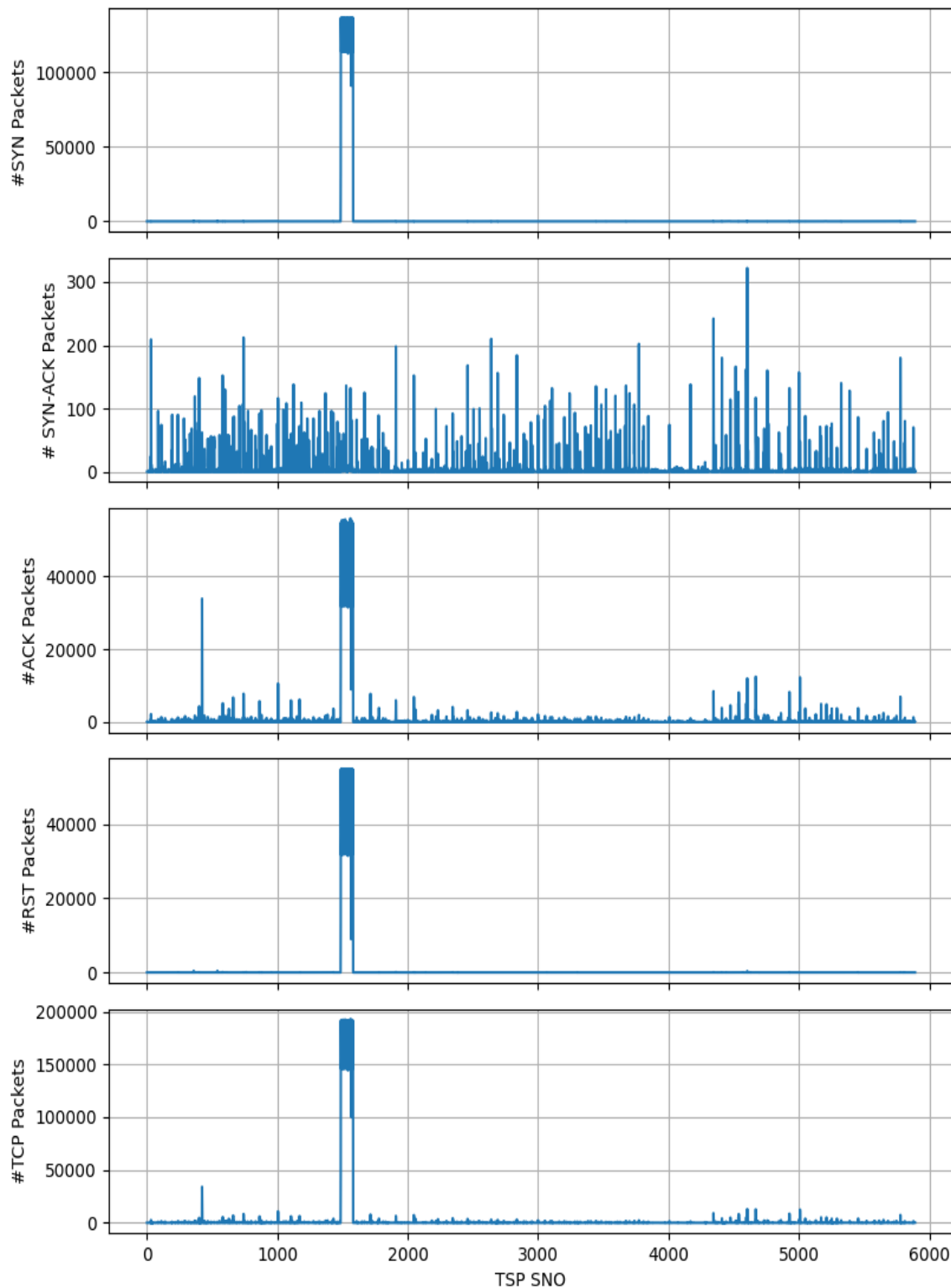


FIGURE 6. Testing set samples.

classification models for assessing attack severity. The performance of these classification models was evaluated using a carefully selected set of metrics, including accuracy,

precision, recall, F1-score, area under the receiver operating characteristic (AUC-ROC) curve, Kappa, and Matthews Correlation Coefficient (MCC). These metrics were chosen

TABLE 3. Performance comparison of different machine learning models for detecting normal and DDoS attack traffic on the training set.

Model	ACCURACY	AUC	RECALL	PREC.	F1	KAPPA	MCC
LGBM	0.9982	0.9998	0.9982	0.9983	0.9982	0.9775	0.9777
GBC	0.9979	0.9997	0.9979	0.998	0.9979	0.9743	0.9747
ADA	0.9974	0.9994	0.9974	0.9976	0.9974	0.9681	0.9687
RF	0.9972	0.9966	0.9972	0.9973	0.9972	0.9647	0.9653
ET	0.9972	0.9966	0.9972	0.9973	0.9972	0.9644	0.965
CATBOOST	0.9969	0.9997	0.9969	0.9971	0.9969	0.9622	0.9628
DT	0.9966	0.9833	0.9966	0.9967	0.9967	0.9581	0.9584
KNN	0.9879	0.9822	0.9879	0.9896	0.9884	0.861	0.8651
LDA	0.9713	0.9502	0.9713	0.9722	0.9637	0.4528	0.5404
RIDGE	0.9711	0.9421	0.9711	0.9719	0.9633	0.4459	0.535
QDA	0.9654	0.8334	0.9654	0.9614	0.9592	0.4128	0.4586
LR	0.6631	0.935	0.6631	0.9552	0.7628	0.1127	0.2213

to provide a comprehensive assessment of the proposed system’s ability to detect and classify DDoS attacks, ensuring its reliability and effectiveness across different scenarios.

The accuracy was used to measure the overall accuracy of the system, offering a straightforward performance evaluation. However, given the potential for imbalanced datasets in DDoS detection, additional metrics are included to provide deeper insights. Precision is critical for minimizing false positives, which could otherwise lead to unnecessary resource consumption and disruptions to legitimate traffic. Recall was vital for ensuring that all actual instances of DDoS attacks were identified, thus preventing undetected threats from escalating. The F1-score balances precision and recall, making it particularly useful in scenarios with imbalanced class distributions, such as those encountered in DDoS traffic analysis.

The AUC-ROC curve provides a graphical representation of the model’s ability to distinguish between classes across various thresholds, highlighting the trade-offs between true-positive and false-positive rates. Kappa and MCC offer robust assessments by accounting for chance agreement and class imbalance, ensuring a more nuanced evaluation of model performance.

In addition, a confusion matrix was employed to provide detailed insights into the classification outcomes, including true positives, false positives, true negatives, and false negatives. This analysis is crucial for refining the model and understanding its behavior when categorizing traffic into low, moderate, and high severity levels.

By leveraging this combination of metrics, the study ensured a holistic assessment of the reliability and suitability of the model for both the detection and severity assessment phases of the proposed system. Each metric complemented the other metrics, offering a clear and robust evaluation framework aligned with the objectives of DDoS attack detection and classification in real-world scenarios.

IV. RESULTS

We tested the proposed system on a Windows 11 system with a 13th-generation Intel Core i5-13500H 64-bit processor at 2.60 GHz and 16 GB of RAM, using PyCaret 3.2.2 and

Python 3.11.0. This setup provided a suitable environment for the execution and evaluation of the method.

We utilized the `setup()` function from the PyCaret library to prepare the dataset for machine learning analysis. This function streamlined the configuration process by automatically handling essential preprocessing tasks, including missing value imputation, categorical variable encoding, and feature scaling using standard scaling. We applied the LightGBM algorithm as part of an iterative imputation strategy for numerical missing values. This ensured accurate estimation by leveraging observed feature relationships. Additionally, normalization was performed to maintain consistency in feature scales, preventing any single feature from disproportionately influencing the model due to its magnitude. To address class imbalances within the dataset, we employed the Synthetic Minority Oversampling Technique (SMOTE) [60] as the resampling method. SMOTE generated synthetic samples for the minority class by interpolating existing data points, thereby balancing the class distribution. This approach reduced bias in model predictions caused by unequal class sizes, ultimately enhancing the reliability and robustness of the machine learning analysis.

PyCaret offered a comprehensive suite of functions designed for efficient model training and evaluation, including `compare_models()`, `create_model()`, `tune_model()`, `evaluate_model()`, `interpret_model()`, and `predict_model`. The `compare_models()` function enabled the comparison of multiple machine learning models on a given training set, facilitating the selection of the best-performing model based on predefined evaluation metrics. The `create_model()` function allowed the instantiation of a specific model for further training and evaluation.

To optimize model performance, `tune_model()` performed hyperparameter tuning using a cross-validated grid search, ensuring systematic fine-tuning for improved outcomes. The `evaluate_model()` function provided an interactive interface to visualize model performance through various plots, including learning curves, ROC curves, confusion matrices, and feature importance charts. Learning curves, in particular, helped assess the model’s training and validation performance over iterations, offering insights into potential issues

TABLE 4. Fine-tuning of LGBM using 10-Fold cross-validation in the first stage.

Fold	ACCURACY	AUC	RECALL	PREC.	F1	KAPPA	MCC
0	0.9974	1.0000	0.9974	0.9976	0.9975	0.9684	0.9688
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0.9974	1.0000	0.9974	0.9976	0.9975	0.9683	0.9688
4	0.9948	0.9993	0.9948	0.9954	0.9950	0.9385	0.9403
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	0.9974	0.9986	0.9974	0.9974	0.9974	0.9664	0.9669
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0.9974	0.9992	0.9974	0.9976	0.9975	0.9683	0.9688
Mean	0.9985	0.9997	0.9985	0.9986	0.9985	0.9810	0.9814
Std	0.0017	0.0005	0.0017	0.0016	0.0017	0.0208	0.0203

such as overfitting or underfitting. The `interpret_model()` function further supported understanding the model's behavior by generating interpretability insights, such as SHAP (SHapley Additive exPlanations) plots, which highlighted the contribution of individual features to the model's predictions.

Finally, the `predict_model()` function generated predictions on the test set, enabling the evaluation of the model's generalization capabilities and its application to unseen data. The proposed methodology consistently adopted this process, ensuring a structured and replicable workflow for model development and performance assessment.

A. STAGE I: DDOS ATTACK DETECTION RESULTS

This study presents an innovative two-stage server health-monitoring system designed to detect and evaluate the severity of distributed denial of service (DDoS) attacks. In the first stage, the system analyzes network traffic data to detect DDoS attacks. By employing a binary classification technique, the system distinguishes between normal and DDoS attack traffic, thereby facilitating recognition of anomalous traffic patterns. Table 3 presents the performance of the different machine learning algorithms in this phase.

1) TRAINING PHASE

The results in Table 3 highlight the performance of various machine-learning models in detecting DDoS attacks during the first stage of the proposed health monitoring system. The LGBM model achieved the highest performance with an accuracy of 99.82%, AUC of 0.9998, and near-perfect scores across all metrics, indicating its exceptional ability to distinguish between normal and attack traffic. GBC followed closely, showing strong metrics, with 99.79% accuracy and an AUC of 0.9997. The ADA and RF models also performed well, with accuracy above 99.7% and robust AUC values.

The ET and CatBoost models exhibited comparable results, maintaining high accuracy and reliability, whereas the DT model achieved slightly lower performance. Models such as the KNN and LDA showed moderate performance, with KNN achieving 98.79% accuracy and LDA at 97.13%. Ridge regression performed similarly to LDA, but both lagged the

top-performing models. The QDA and LR models exhibited relatively lower accuracies, at 96.54% and 66.31%, respectively, with reduced kappa and MCC values. Overall, the LGBM emerged as the most effective model for accurate and reliable DDoS attack detection in this phase.

Figure 7 illustrates the learning curves for the machine learning models in Stage I, highlighting their training and validation performance. These curves help assess model generalization, detecting potential underfitting or overfitting. The learning curve (Figure 7(a)) for LightGBM (LGBM) demonstrated rapid convergence with minimal variance, indicating excellent generalization and stability. Similarly, the GBC model showed smooth and consistent learning (Figure 7(b)), with a negligible gap between the training and validation scores, reflecting its robust performance. The ADA model also maintained a steady improvement in accuracy, balancing the training and validation scores effectively (Figure 7(c)). The RF model exhibited quick initial learning, followed by a plateau at high accuracy levels, with minimal differences between training and validation, confirming its reliability (Figure 7(d)).

The ET model mirrors the RF model, achieving stable performance with efficient learning and strong generalization (Figure 7(e)). The CatBoost model showed efficient convergence with minimal gaps between training and validation accuracy, highlighting balanced learning behavior (Figure 7(f)). On the other hand, the DT model (Figure 7(g)) overfits quickly, as is evident from the high training accuracy compared to the lower validation accuracy, indicating weaker generalization. The KNN model achieved good initial accuracy (Figure 7(h)), but a visible gap between the training and validation scores suggests sensitivity to the dataset. The LDA model displayed steady progress with slower convergence, achieving moderate accuracy, which reflects its limited complexity in capturing patterns (Figure 7(i)). RIDGE demonstrated stable improvement with minimal variance, although it remained below the top performing models in terms of overall accuracy (Figure 7(j)).

The learning curve (Figure 7(k)) for the QDA model shows a sharp decline in the training score, whereas the

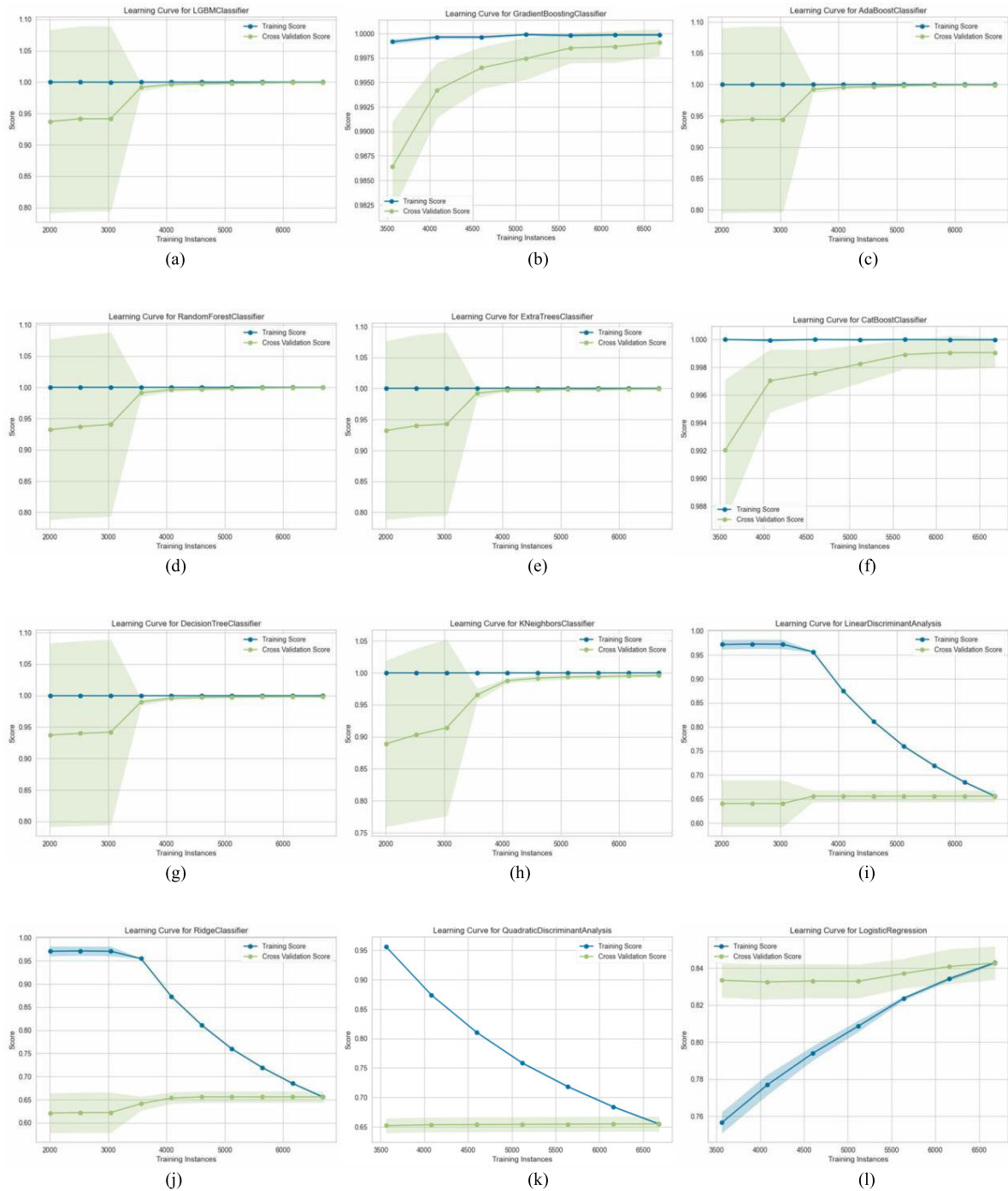


FIGURE 7. Learning curves of machine-learning models for detecting normal and DDoS attack traffic: (a)LGBM, (b)GBC, (c)ADA, (d)RF, (e)ET,(f) CATBOOST,(g)DT, (h)KNN, (i)LDA, (j)RIDGE, (k)QDA, (l)LR.

cross-validation score remains low and stable, indicating poor generalization. The model struggles to learn from increasing amounts of data, making it ineffective for classifying anomalous network traffic.

Similarly, the learning curve (Figure 7(l)) for the LR model shows a lower accuracy and a persistent gap, highlighting its challenges in adapting to the complexity of the dataset. Overall, the learning curves emphasize the strengths

of ensemble methods, such as LGBM and GBC, which demonstrate superior generalization and adaptability compared to other models. Following its initial implementation, we fine-tuned the LGBM using a 10-fold cross-validation method, as presented in Table 4. These results detail the model's performance, where the accuracy scores varied from 0.9948 to 1 across the ten folds, showing minimal variability. Despite slight fluctuations, the average accuracy maintained

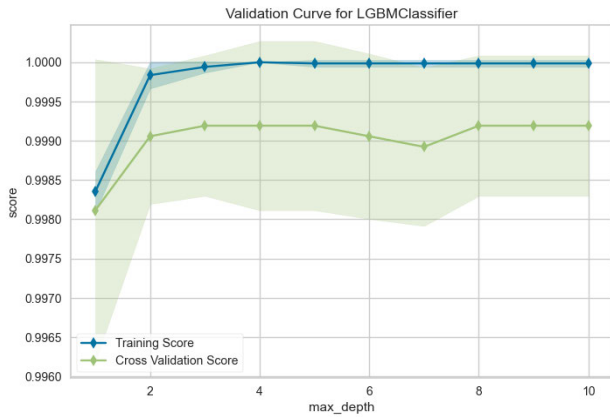


FIGURE 8. Validation curve for the light gradient boosting classifier.

a robust value of 0.9985, indicating consistent and strong performance. With a standard deviation of 0.0017, the results affirm the model's stability and reliability in predicting server health and detecting distributed denial of service (DDoS) attacks.

As the value of "actual_estimator_max_depth" increases, as shown in Figure 8, the validation curve for the LGBM and both the training and cross-validation scores improved. This trend suggests that increasing the maximum depth of individual estimators in the gradient-boosting model leads to enhanced model performance. Figure 9 shows the ROC curve for the LGBM model. With an AUC of 1.00 for both Class 0 (normal traffic) and Class 1 (DDoS attack traffic), along with the micro- and macro-average ROC curves, the GBC model was very good at describing the difference between classes and classifying everything correctly. This underscores the robustness and effectiveness of binary classification tasks.

2) TESTING PHASE

Table 5 summarizes the performance metrics of LGBM for the test set. An accuracy score of 0.9988 indicated the proportion of correctly classified instances. An AUC value of 0.9999 signifies the classifier's ability to distinguish between the positive and negative instances. The recall rate of 0.9988 suggests that LGBM effectively identifies the most true positive instances, whereas a precision of 0.9988 indicates a high proportion of true positive predictions out of all positive predictions. An F1 score of 0.9988 represents the harmonic mean of the precision and recall. Additionally, Cohen's kappa coefficient of 0.9907 and MCC of 0.9907 demonstrated substantial agreement between the predicted and actual classifications, considering both true positives, false positives, and false negatives. Overall, these metrics highlight the strong performance of the light-gradient boosting machine on the testing data during this phase.

B. STAGE II: ATTACK SEVERITY ASSESSMENT RESULTS

In the second stage of the proposed system, machine learning algorithms were used to assess the severity of detected DDoS

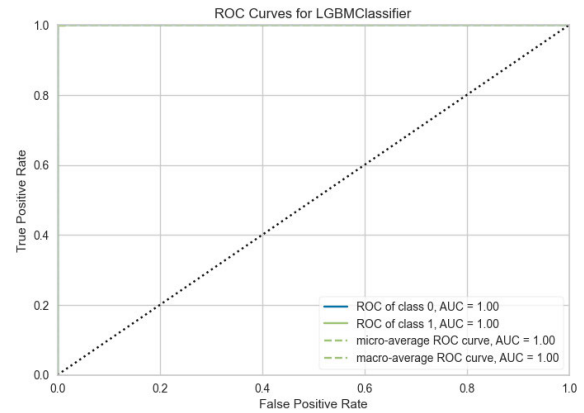


FIGURE 9. ROC curve for the light gradient boosting classifier.

attacks. Multiclass classification techniques were applied to categorize DDoS traffic into three severity levels: low, moderate, and high.

1) TRAINING PHASE

Table 6 presents the classification performance of various models in detecting SYN Flood DDoS attacks across normal, low, moderate, and high severity levels. LGBM achieved the highest accuracy of 0.9985, with near-perfect recall (0.9985), precision (0.9983), and F1-score (0.9983). Its Kappa (0.9808) and MCC (0.981) values indicated strong model reliability. RF and CatBoost closely followed the accuracy of 0.9984, displaying almost identical precision-recall values. CatBoost recorded the highest AUC of 0.8, suggesting excellent discriminatory power across categories. DT performed slightly below these models, with an accuracy of 0.9982 and an AUC of 0.7932, showing effective classification but slightly reduced generalization.

GBC achieved an accuracy of 0.9979, but its AUC of 0.7 was notably lower than that of the other ensemble methods, indicating a potential weakness in distinguishing certain categories.

ET performed similarly to GBC in terms of accuracy (0.9979) but showed a higher AUC of 0.7999, suggesting improved classification ability. KNN achieved an accuracy of 0.9902, though its AUC (0.7927) was slightly lower, indicating that its classification effectiveness was slightly inferior compared to the tree-based models.

QDA exhibited moderate performance, with an accuracy of 0.9718 and an AUC of 0.7666, demonstrating its capability to differentiate between traffic types but with lagging ensemble methods. LR and LDA showed limited effectiveness, with LR achieving 0.6324 accuracy and LDA dropping to 0.3748. Despite their high precision (~ 0.97), both models suffered from low recall and F1 scores, which reduced their overall reliability. Ridge performed similarly to LDA, with an accuracy of 0.3642 and an AUC of 0.2, indicating a poor classification capability.

ADA had the weakest performance among all models, with an accuracy of 0.0393 and extremely low recall, precision, and F1-score, making it unsuitable for this classification task.

TABLE 5. LGBM classifier performance on the testing set in the first stage.

ACCURACY	AUC	RECALL	PREC.	F1	KAPPA	MCC
0.9988	0.9999	0.9988	0.9988	0.9988	0.9907	0.9907

TABLE 6. Classification performance of different models for detecting SYN flood DDoS attacks across normal, Low, Moderate, and High severity levels on the training set.

Model	ACCURACY	AUC	RECALL	PREC.	F1	KAPPA	MCC
LGBM	0.9985	0.7991	0.9985	0.9983	0.9983	0.9808	0.981
RF	0.9984	0.7999	0.9984	0.9985	0.9985	0.9808	0.981
CATBOOST	0.9984	0.8	0.9984	0.9985	0.9985	0.981	0.9812
DT	0.9982	0.7932	0.9982	0.9983	0.9982	0.9779	0.9782
GBC	0.9979	0.7000	0.9979	0.9981	0.9979	0.9743	0.9748
ET	0.9979	0.7999	0.9979	0.9981	0.998	0.9745	0.9749
KNN	0.9902	0.7927	0.9902	0.9918	0.9907	0.8869	0.89
QDA	0.9718	0.7666	0.9718	0.9838	0.9759	0.7325	0.7539
LR	0.6324	0.5000	0.6324	0.9674	0.7483	0.1147	0.2401
LDA	0.3748	0.2000	0.3748	0.9708	0.51	0.0532	0.1823
RIDGE	0.3642	0.2000	0.3642	0.9706	0.4981	0.0513	0.1798
ADA	0.0393	0.2000	0.0393	0.0098	0.0113	0.0111	0.2423

Figure 10 illustrates the learning curves of the 12 machine-learning models for the multiclass classification of SYN Flood DDoS attack network traffic into normal, low, moderate, and high severity levels. The LGBM learning curve (Figure 7(a)) shows excellent performance, with training and cross-validation scores converging at high levels, indicating strong generalization. The GBC model (Figure 7(b)) also achieved consistent results with a slight gap between the training and cross-validation scores. The ADA model (Figure 7(c)) displays fluctuations in cross-validation performance, reflecting the sensitivity to dataset size and challenges in generalization. The RF (Figure 7(d)) and ET (Figure 7(e)) models exhibited stable and robust performance with closely aligned training and cross-validation curves, confirming effective learning.

The CatBoost (Figure 7(f)) model achieved smooth convergence with minimal gaps, demonstrating strong adaptability. The DT model (Figure 7(g)) showed a small gap between the training and cross-validation scores, indicating a slight tendency to overfit. The KNN model (Figure 7(h)) improved steadily as more data were added, with training and cross-validation scores aligning well, showing strong adaptability. The LDA model (Figure 7(i)) initially showed a gap between the training and cross-validation scores, but the performance stabilized with increased data. The RIDGE model (Figure 7(j)) experienced significant fluctuations in scores, reflecting sensitivity to data distribution, although it converged to moderate accuracy.

The QDA model (Figure 7(k)) demonstrated high training accuracy but a notable gap with cross-validation scores, suggesting overfitting. The LR model (Figure 7(l)) shows gradual improvement in cross-validation scores with additional data, although a consistent gap with training scores indicates limited generalization.

After the initial implementation, the LGBM was fine-tuned using a 10-fold cross-validation method. Table 7 presents the results. These results demonstrate the LGBM model's performance after fine-tuning, displaying accuracy scores ranging from 0.9948 to 1.0000 across the ten folds, with a mean accuracy of 0.9990. This consistency reflects the model's robustness across diverse data subsets. Moreover, the small standard deviation of 0.0017 emphasizes the stability and reliability of the GBC model, indicating minimal variability in its performance.

As shown in Figure 11, the validation curve for the LGBM Classifier illustrates the performance of the model with respect to the max_depth hyperparameter. As max_depth increased from one to two, there was a significant improvement in both the training and cross-validation scores, indicating a better fit with a slightly deeper tree. Beyond max_depth 2, the training score approached 1.0, suggesting a near-perfect fit to the training data, while the cross-validation score stabilized close to the training score, ranging from 0.9985 to 0.9990, respectively.

This indicates that the model effectively balances the bias and variance with a maximum depth of approximately 2–4, providing a strong fit without overfitting. The minimal gap between the training and cross-validation scores, along with the stable cross-validation score at higher max_depth values, suggests that the model generalizes well to unseen data with minimal variability, confirming its robustness across different max_depth settings.

Figure 12 shows the ROC curve for an LGBM classifier, which shows how well it sorts traffic into four groups: high severity (class 0), low severity (class 1), moderate severity (class 2), or normal (class 3). The High and Moderate severity classes both achieved perfect discrimination with an AUCs of 1.00, whereas the Low severity class showed slightly



FIGURE 10. Learning curves for multiclass classification of network traffic: Normal, Low, Moderate, and High Severity Categories (a)LGBM, (b)GBC, (c)ADA, (d)RF, (e)ET, (f)CATBOOST, (g)DT, (h)KNN, (i)LDA, (j)RIDGE, (k)QDA, (l)LR.

lower performance with an AUC of 0.98. With an AUC of 0.99, the normal traffic class demonstrated near-perfect performance.

The micro-average ROC curve, with an AUC of 1.00, reflects excellent overall performance, whereas the macro-average ROC curve, with an AUC of 0.99, indicates a consistently strong classification across all classes. These results demonstrate that the LGBM classifier is highly

effective in distinguishing between different levels of attack severity and normal traffic.

The SHAP summary plot in Figure 13 offers an interpretation of the LGBM Classifier's decision making in categorizing network traffic into high, low, moderate, and normal categories. Key features such as the number of ACK-Packets, SYN-Packets, RST-Packets, SYN-ACK-Packets, and TCP-Packets primarily determine the model output. In an

TABLE 7. Fine-tuning of LGBM using 10-Fold cross-validation in the second stage.

Fold	ACCURACY	AUC	RECALL	PREC.	F1	KAPPA	MCC
0	0.9974	0.9892	0.9974	0.9949	0.9961	0.9667	0.9673
1	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0.9948	0.9995	0.9948	0.9948	0.9948	0.9354	0.9354
6	0.9974	1.0000	0.9974	0.9976	0.9975	0.9687	0.9692
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Mean	0.9990	0.7989	0.9990	0.9987	0.9988	0.9871	0.9872
Std	0.0017	0.3994	0.0017	0.0021	0.0019	0.0214	0.0213

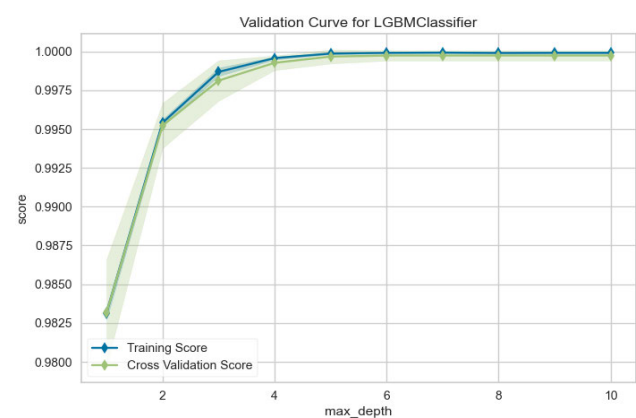


FIGURE 11. Validation curve for light gradient boosting classifier.

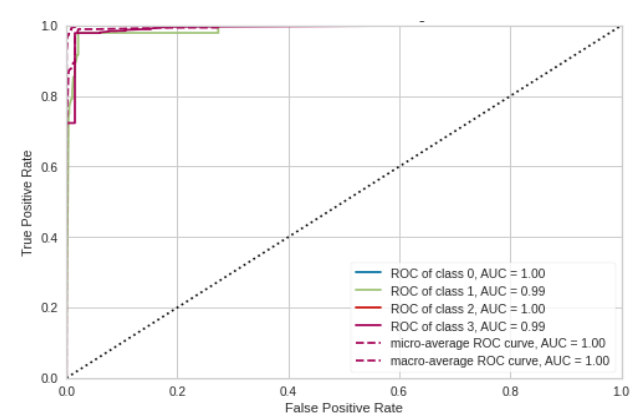


FIGURE 12. ROC curve for light gradient boosting classifier.

individual prediction, each dot represents a SHAP value for a feature, with the color indicating the feature value (red for high, blue for low). The plot reveals how specific packet counts influence the model’s predictions, highlighting the importance of these features in determining the traffic classification. This interpretability is essential for understanding the behavior of the model in terms of network security and anomaly detection.

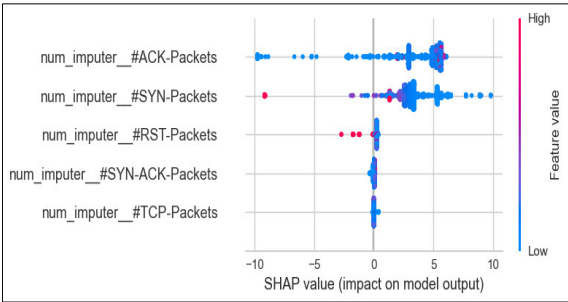


FIGURE 13. SHAP summary plot for feature impact on LGBM classifier.

Among these features, SYN-Packets and ACK-Packets have the most pronounced impact on the model’s decisions. High values of SYN-Packets (represented in red) significantly increase the SHAP values, guiding the model to classify traffic as “High Severity level”. This reflects scenarios such as SYN flood attacks, where a large number of SYN packets indicate excessive connection initiation attempts, which are characteristic of malicious activities. Conversely, low values of SYN-Packets (in blue) reduce the SHAP values, steering the model towards predictions like “Normal” or “Low Severity level,” as such counts typically align with benign traffic patterns.

In contrast, the ACK-Packets feature indicates regular network communications. High counts of ACK packets (red) result in low or negative SHAP values, directing the model towards “Normal” traffic classification. This is because high ACK-Packet counts correspond to acknowledgment-heavy and stable communication patterns, which are characteristic of normal network behavior. Conversely, low counts of ACK-Packets (blue) positively influence SHAP values, leading the model to predict “Low” or “Moderate” traffic, where acknowledgment activity may be less frequent or irregular.

The interaction between the SYN and ACK Packets provides key insights into the interpretability of the model. SYN-Packets signal potential anomalies, whereas ACK-Packets reflect the communication stability. These features enable effective traffic classification into Normal, Low, Moderate, and High severity levels, thereby enhancing the transparency

TABLE 8. Performance comparison of different machine learning models on the test set for detecting network traffic across normal, Low, Moderate, and High severity levels.

Model	ACCURACY	AUC	RECALL	PREC.	F1	KAPPA	MCC
GBC	0.9994	1	0.9994	0.9994	0.9994	0.9992	0.9992
ET	0.9886	1	0.9886	0.989	0.9886	0.9848	0.9849
KNN	0.987	0.9937	0.987	0.9872	0.987	0.9827	0.9827
RF	0.9866	1	0.9866	0.9872	0.9866	0.9821	0.9823
CATBOOST	0.9862	1	0.9862	0.9868	0.9862	0.9816	0.9818
LGBC	0.9851	1	0.9851	0.9858	0.9851	0.9801	0.9804
DT	0.9841	0.9894	0.9841	0.9849	0.9841	0.9788	0.9791
LR	0.8716	0.9949	0.8716	0.9014	0.865	0.8287	0.8429
LDA	0.7792	0.9937	0.7792	0.8544	0.735	0.7056	0.7493
RIDGE	0.759	0.6333	0.759	0.8168	0.7167	0.6787	0.7202
QDA	0.7552	0.7938	0.7552	0.8523	0.7101	0.6735	0.7177
ADA	0.5	0.4167	0.5	0.25	0.3333	0.3333	0.4082

of the LGBM Classifier for server health prediction and network management.

2) TESTING PHASE

The test set used for this study initially exhibited significant class imbalance, with the majority class (Normal) containing 5,479 samples, while the minority classes (“Low”, “Moderate” and “High”) had only 308, 11, and 94 samples, respectively. To address this issue and ensure fair evaluation, the Synthetic Minority Oversampling Technique (SMOTE) [54] is applied. This technique generated synthetic samples for minority classes, resulting in a perfectly balanced dataset in which each class contained 5,479 samples. This preprocessing step eliminated the bias caused by class imbalance and provided an equal representation of all classes, facilitating a robust comparison of the model performance. Table 8 summarizes the performance of the machine-learning models for the test set. Among these, the GBC model achieved the highest performance, with an accuracy of 99.94%, AUC of 1.0, and F1 score of 0.9994. It demonstrated nearly perfect precision and recall, both at 0.9994, along with a Kappa and MCC of 0.9992, highlighting its robustness and reliability.

Other ensemble models, such as ET and RF, also performed exceptionally well. ET achieved an accuracy of 98.86%, an AUC of 1.0, and an F1 score of 0.9886, whereas RF achieved an accuracy of 98.66%, an AUC of 1.0, and an F1 score of 0.9866. Both models showed strong agreement, with Kappa and MCC values exceeding 0.98. The KNN model also demonstrated a high performance, with an accuracy of 98.7% and an F1 score of 0.987. The CatBoost and LightGBM models performed comparably, achieving accuracy of 98.62% and 98.51%, respectively. The DT model also showed reliable performance, with an accuracy of 98.41% and an AUC of 0.9894. The LR model achieved moderate performance, with an accuracy of 87.16%, an AUC of 0.9949, and a relatively high precision of 0.9014, but with lower F1 and Kappa values, indicating challenges in capturing complex patterns in the dataset.

The LDA and RIDGE models achieved accuracies of 77.92% and 75.9%, respectively, with moderate performances across the other metrics. The QDA model performed slightly better than the ridge model, with an accuracy of 75.52%. In contrast, the ADA model showed the weakest performance, with an accuracy of 50%, F1 score of 0.3333, and MCC of 0.4082, indicating its unsuitability for this task.

Figure 14 illustrates the performance of various models on the test set evaluated using confusion matrices to assess their prediction accuracy across all classes.

The confusion matrix for the LGBM model is shown in Figure 14 (a). The model classified network traffic into four categories: Normal, Low, Moderate, and High attack severity. The Normal class was accurately classified 5465 times, with minor misclassifications—11 instances predicted as Low and 3 as Moderate. The Low severity class was also accurately classified, correctly predicting 5475 instances and misclassifying only 4 instances as Normal. For the Moderate severity class, the model correctly identified 5170 instances, with 309 cases incorrectly labeled as High severity. The High severity class achieved perfect classification, correctly identifying all 5479 instances.

Figure 14 (b) shows the confusion matrix of the Gradient Boosting Classifier (GBC). The model accurately classified 5470 Normal instances, misclassifying only 9 instances as Low.

It also correctly predicted 5475 Low severity instances, with only 4 instances misclassified as Normal. The Moderate and High severity classes were classified perfectly, each with all 5479 instances correctly predicted and no errors. Overall, the GBC achieved excellent classification accuracy.

Figure 14 (c) shows the confusion matrix of the AdaBoost Classifier (ADA). The model accurately classified 5478 Normal instances, misclassifying only 1 instance as Low. However, it misclassified all 5479 Low severity instances as Normal. The model also correctly classified all 5479 Moderate severity instances but incorrectly predicted all 5479 High severity instances as Moderate. Overall,

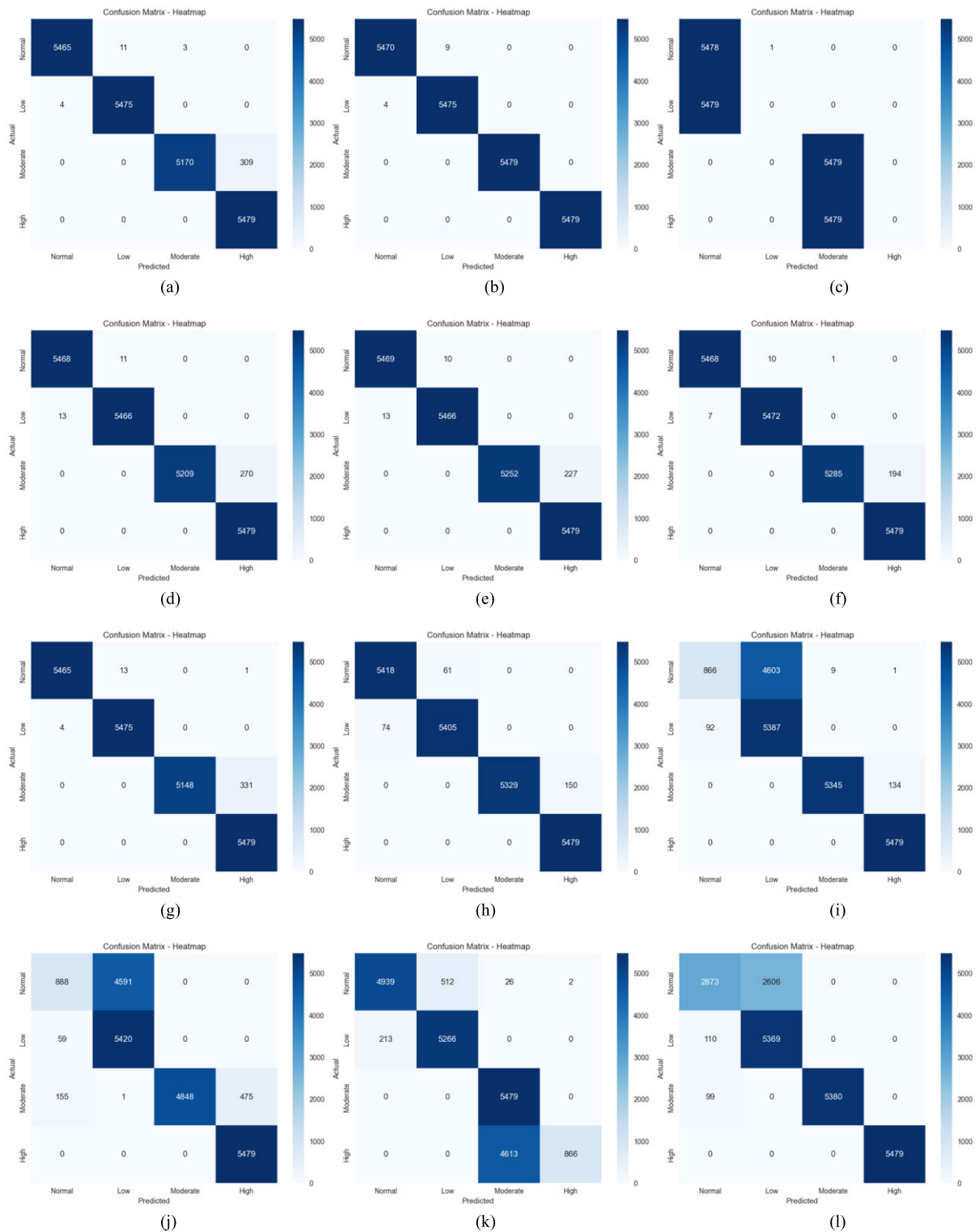


FIGURE 14. Confusion matrices for multiclass classification of network traffic: Normal, Low, Moderate, and High Severity Categories: (a) LGBM, (b) GBC, (c) ADA, (d) RF, (e) ET, (f) CATBOOST, (g) DT, (h) KNN, (i) LDA, (j) RIDGE, (k) QDA, (l) LR.

the ADA classifier struggled significantly in distinguishing between Normal-Low and Moderate-High severity levels.

Figure 14 (d) shows the confusion matrix of the Random Forest (RF) classifier. The RF accurately classified

5468 Normal instances, misclassifying only 11 as Low. The Low severity class was predicted correctly 5466 times, with 13 instances incorrectly labeled as Normal. Moderate severity instances were correctly classified 5209 times, though

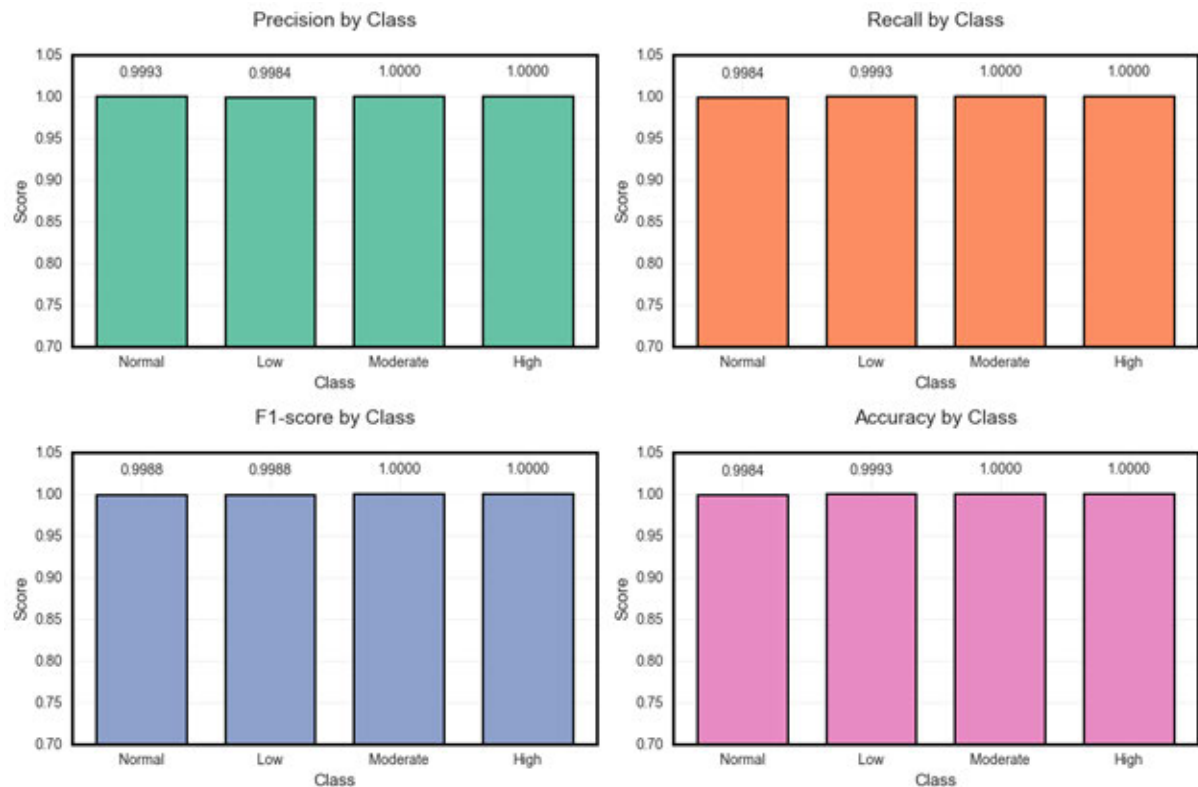


FIGURE 15. Class-wise performance on testing set: GBC Precision, Recall, F1-Score, Accuracy.

270 were misclassified as High. The High severity class was perfectly classified, with all 5479 instances identified correctly.

Figure 14 (e) shows the confusion matrix of the Extra Trees (ET) classifier. The ET model accurately classified 5469 Normal instances with only 10 misclassified as Low. The Low severity class was correctly classified 5466 times, with 13 instances misclassified as Normal. For Moderate severity, 5252 instances were correctly identified, while 227 were incorrectly predicted as High. The High severity class was perfectly classified, correctly identifying all 5479 instances.

Figure 14 (f) shows the confusion matrix of the Decision Tree (DT) classifier. The DT model correctly classified 5465 Normal instances, with minor errors—13 misclassified as Low and 1 as High. It accurately identified 5475 Low severity instances, misclassifying only 4 instances as Normal. The Moderate severity class had 5148 correct predictions, but 331 instances were misclassified as High. The High severity class was classified perfectly with all 5479 instances correctly predicted.

Figure 14 (g) shows the confusion matrix of the K-Nearest Neighbors (KNN) classifier. The model correctly classified 5418 Normal instances, misclassifying 61 as Low. The Low severity class was accurately classified 5405 times, with 74 instances wrongly labeled as Normal. For the Moderate severity level, 5329 instances were correctly identified, but 150 instances were misclassified as High. The High

severity class was perfectly classified, correctly identifying all 5479 instances.

Figure 14 (h) shows the confusion matrix of the Linear Discriminant Analysis (LDA) classifier. The model had significant misclassification issues between Normal and Low severity levels, correctly classifying only 866 Normal instances and misclassifying 4603 as Low. For the Low severity class, 5387 instances were correctly identified, with 92 incorrectly predicted as Normal. Moderate severity had 5345 correct predictions but misclassified 134 instances as High. High severity was accurately classified in all 5479 cases.

Figure 14 (i) shows the confusion matrix of the Ridge classifier. The model accurately classified only 888 Normal instances, while a significant 4591 were misclassified as Low. The Low severity class had 5420 correctly predicted instances, with 59 errors classified as Normal. The Moderate severity class was predicted accurately 4848 times, but 134 cases were misclassified as High, and 59 as Low. The High severity class was classified perfectly, with all 5479 instances correctly identified. The results show that the Ridge model had notable difficulty distinguishing between Normal and Low severity classes.

Figure 14 (j) shows the confusion matrix of the Quadratic Discriminant Analysis (QDA) classifier. The model accurately classified 4939 Normal instances but misclassified

512 instances as Low, 26 as Moderate, and 2 as High. The Low severity class was correctly classified 5266 times, with 213 instances wrongly labeled as Normal. The Moderate class was perfectly classified, correctly identifying all 5479 instances. However, the model significantly misclassified the High severity class, correctly predicting only 866 instances and misclassifying 4613 as Moderate. The QDA model showed notable confusion between Moderate and High severity classes.

Figure 14 (k) shows the confusion matrix of the Logistic Regression (LR) classifier. The model correctly identified only 2873 Normal instances, with a large misclassification of 2606 instances as Low. The Low severity class had 5369 correct classifications but 110 misclassifications as Normal. The Moderate severity class was accurately classified 5380 times, misclassifying 99 instances as Normal. The High severity class was classified perfectly, with all 5479 instances correctly identified. The LR classifier had significant difficulties distinguishing between Normal and Low severity classes.

The Gradient Boosting Classifier showed excellent performance in classifying network traffic into “Normal,” “Low,” “Moderate,” and “High” severity levels, achieving near-perfect accuracy. Figure 15 demonstrates its strong capability to minimize misclassifications with high precision and recall, indicating its suitability for identifying DDoS-related traffic. Additionally, the model effectively differentiated severity levels of SYN Flood DDoS attacks, providing valuable insights into their impact. These results confirm the robustness and relevance of the Gradient Boosting Classifier in this research, highlighting its potential to enhance DDoS attack detection and severity assessment.

V. DISCUSSION

In the current digital era, safeguarding server infrastructure against DDoS attacks is essential for maintaining uninterrupted online operations. To ensure server health, this study introduces a novel two-stage monitoring system designed to classify the severity of SYN Flood attacks, a critical type of DDoS attack. The system addresses significant gaps in existing research and leverages the TSD-DDoS dataset derived from the benchmark CIC-DDoS2019 dataset. The TSD-DDoS dataset was specifically prepared to facilitate time-series modeling, thereby enabling enhanced resource allocation and detection strategies for DDoS attacks.

Several studies [48], [49], [50], [51], [52], [53], [54], [55], [56] have explored DDoS attack detection using machine learning and deep learning techniques. These works have successfully implemented various models for identifying malicious traffic and mitigating attacks in different network environments, including SDN, IoT, and Industry 5.0. However, while these approaches effectively classify network traffic as normal or malicious, they do not assess the severity of attacks. Unlike these methods, the proposed system not only detects DDoS attacks but also categorizes them into

different severity levels (low, moderate, and high), enabling more informed and effective mitigation strategies. This additional capability enhances the practicality of DDoS detection systems by allowing cybersecurity professionals to prioritize responses based on attack intensity, making the proposed method more comprehensive in handling DDoS threats.

The proposed system demonstrated outstanding performance at both stages of operation. In the first stage, LGBM achieved impressive accuracy of 0.9988, effectively identifying SYN Flood traffic patterns from normal network activity. This underscores the reliability of the system in promptly detecting malicious activities. In the second stage, the system employs a multiclass classification approach to assess the severity of the detected SYN Flood attack, categorizing them into low, medium, and high severity levels. In this stage GBC performed exceptionally, achieving an accuracy of 0.9994 and enabling cybersecurity professionals to prioritize and implement appropriate mitigation strategies.

The proposed system is designed to study the impact of SYN Flood DDoS attacks on server health by defining normal traffic and categorizing attack severities into low, medium, and high levels using specific constants ($a = 1$ for low, $b = 3$ for medium, and $c = 6$ for high severe attacks) as outlined in Equation 3. These constants are derived from traffic speed thresholds of up to 20 Gbps and calibrated using the TSD-DDoS dataset. The fine-tuning capability of these constants ensures the system's adaptability to varying network conditions, enabling it to handle both low- and high-speed SYN Flood attacks effectively. This flexibility highlights the potential of the proposed method to function in diverse network environments and with varying traffic loads.

Although the proposed system has not yet been deployed or tested in a live environment, the fine-tuning capability of these constants demonstrates the method's potential adaptability for real-world applications. The flexibility to calibrate the system according to specific traffic speeds ensures its readiness to handle diverse network environments and varying traffic loads, paving the way for future deployment in live scenarios.

The system currently focuses on SYN Flood attacks, the methodology can serve as a reference template for addressing other types of DDoS attacks. By defining specific equations tailored to the normal traffic and severity levels for different attack types, the proposed approach can be extended to manage a broader spectrum of DDoS attack detection. For instance, reflective- or amplification-based DDoS attacks can be addressed by creating new constants and equations to account for their unique characteristics and traffic patterns. This adaptability makes the framework versatile and scalable and provides a foundation for further research and development.

The generalizability of the proposed system to other types of DDoS attacks like other DDoS variants, such as UDP floods or amplification attacks or varying network

environments is a potential limitation that warrants further investigation. Different DDoS attack types may exhibit unique traffic patterns or behaviors, necessitating a customized calibration of the system's parameters. Furthermore, the scalability and latency of the system under live, large-scale network conditions have not yet been tested. While the current framework shows promise, future work will focus on running machine learning models in parallel to enhance scalability and make the system better suited for live environments. Parallel processing can significantly reduce latency, allowing the system to handle high traffic loads efficiently and respond in real-time.

Despite these limitations, this study makes a significant contribution to the field of cybersecurity by addressing the key shortcomings of previous studies. Unlike traditional approaches that rely on outdated datasets or are limited to binary classification, the proposed system integrates severity assessment and leverages the TSD-DDoS dataset, which captures modern traffic patterns and temporal trends. These features enhance the system's applicability to real-world scenarios, whereas its specific focus on SYN Flood attack severities provides critical insights to bolster server health and resilience.

VI. CONCLUSION AND FUTURE WORK

The proposed two-stage server health monitoring system provides a robust and adaptable solution for detecting and classifying SYN Flood DDoS attacks based on severity. By leveraging the TSD-DDoS dataset, the system effectively captures temporal traffic patterns, improving its detection capabilities. The experimental results demonstrate high accuracy, with LightGBM (LGBM) achieving the best accuracy of 0.9988 in Stage I (DDoS Attack Detection) and Gradient Boosting Classifier (GBC) achieving the highest accuracy of 0.9994 in Stage II (Attack Severity Assessment). These results highlight the system's reliability and effectiveness in identifying and classifying attacks.

Understanding attack severity levels is essential for cybersecurity professionals, as it allows them to prioritize mitigation efforts based on attack intensity. Unlike traditional studies that rely on binary classification to distinguish normal and DDoS traffic, this study incorporates severity assessment, enabling a more proactive and efficient response to DDoS threats. This approach strengthens DDoS mitigation strategies, enhancing network resilience against evolving cyber threats.

Future work will focus on enhancing accuracy and scalability by incorporating stacking machine learning techniques and advanced time-series forecasting models for predicting attack severity. Additionally, parallel execution of machine learning models will be explored to optimize the system for real-time deployment, ensuring efficient detection in high-traffic environments. These improvements will further enhance the system's adaptability and practicality for real-world cybersecurity applications.

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